



Forward dynamics simulation of a simplified neuromuscular-skeletal-exoskeletal model based on the CMA-ES optimization algorithm: framework and case studies

Wei Jin^{1,2} · Jiaqi Liu³ · Qiwei Zhang^{1,2} · Xiaoxu Zhang^{1,2} · Qining Wang⁴ · Jian Xu³ · Hongbin Fang^{1,2}

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Abstract

The modeling and simulation of coupled neuromusculoskeletal-exoskeletal systems play a crucial role in human biomechanical analysis, as well as in the design and control of exoskeletons. This study incorporates the integration of exoskeleton models into a reflex-based gait model, emphasizing human-exoskeleton interaction. Specifically, we introduce an optimization-based dynamic simulation framework that integrates a neuromusculoskeletal feedback loop, multibody dynamics, human-exoskeleton interaction, and foot-ground contact. The framework advances in human-exoskeleton interaction and muscle reflex model refinement. Without relying on experimental measurements or empirical data, our framework employs a stepwise optimization process to determine muscle reflex parameters, taking into account multidimensional criteria. This allows the framework to generate a full range of kinematic and biomechanical signals, including muscle activations, muscle forces, joint torques, etc., which are typically challenging to measure experimentally. To evaluate the validity of the framework, we compare the simulated results with experimental data obtained from a healthy subject wearing an exoskeleton while walking at different speeds (0.9, 1.0, and 1.1 m/s) and terrains (flat and uphill). The results demonstrate that our framework can capture the qualitative differences in muscle activity associated with different functions, as well as the evolutionary patterns of muscle activity and kinematic signals with respect to varying walking conditions, with the Pearson correlation coefficient $R > 0.7$. Simulations of the human walking with the exoskeleton in both passive mode and assisting mode at a peak torque of 20 N·m are further conducted to investigate the effect of exoskeleton assistance on human biomechanics. The simulation framework we propose has the potential to facilitate gait analysis and performance evaluation of coupled human-exoskeleton systems, as well as enable efficient and cost-effective testing of novel exoskeleton designs and control strategies.

Keywords Neuromusculoskeletal modeling · Musculoskeletal dynamics · Human-exoskeleton interaction · Human biomechanics · Hip exoskeleton

1 Introduction

Wearable exoskeletons have garnered considerable attention in recent years due to their potential in facilitating human movement-related applications, making them a focal point of biomechanical and robotic research [1–3]. For example, gait rehabilitation robots have demonstrated superior effectiveness compared to conventional treatments in restoring mobility for patients with spinal cord injuries [4]. Passive exoskeletons have been developed to mitigate the risk of strain-related injuries in the workplace [5] and reduce energy expenditure during movement [6]. On the other hand, powered exoskeletons are designed to provide active assistance in movement for healthy individuals and those with motor impairments [7–10]. Their applications encompass tasks such as aiding individuals in carrying heavy objects over long distances [11], facilitating long-term rehabilitation, supporting the mobility of injured or elderly individuals [12], etc. Despite their immense application potential, the design, control, and performance evaluation of wearable exoskeletons present significant challenges due to several reasons. Firstly, wearable exoskeletons are closely coupled to the human body and therefore necessitate stringent safety requirements, as even minor malfunctions can have severe consequences. Secondly, evaluating the performance of exoskeletons, including aspects like reliability, human-machine interaction, and fault tolerance, etc., demands costly prototyping processes and laborious experiments involving human subjects. Lastly, the diversity of environmental and task conditions further aggravates the burden of experimental research.

Given the significant time, labor, and economic costs for experimental studies, conducting dynamic simulations becomes a viable alternative for evaluating exoskeleton performance [13–15]. Developing a dynamic simulation framework that integrates both the human skeleton and the exoskeleton would enable the estimation of the kinematics of the coupled human-exoskeleton system outside the laboratory setting [16]. This framework facilitates effective analysis of the effects of exoskeletons on human motions and enables convenient assessment of the level of human-exoskeleton interaction. Moreover, dynamic simulations eliminate the need for frequent fabrication of new exoskeleton prototypes and costly experiments, making it possible to test new or improved exoskeleton designs and control strategies at low costs to enhance performance [17, 18]. Additionally, by incorporating the human neuromusculoskeletal system into the dynamic simulation framework, it becomes feasible to perform human gait analysis and obtain valuable information on joint torques, muscle forces, and muscle activations that are challenging to measure experimentally [19]; such information is crucial for analyzing the assistance provided by exoskeletons in human movement. Furthermore, by incorporating tunable human-environment interaction within the dynamic simulation framework, it becomes possible to evaluate the performance of human-exoskeleton coupled systems across different tasks and environments. However, the existing dynamic models and simulation frameworks have not comprehensively addressed all these factors, thereby limiting the widespread application of dynamic simulations in exoskeleton design and control.

Researchers have extensively investigated the intricate issue of musculoskeletal-exoskeletal multibody simulation, employing various methods such as inverse, forward, or a combination of both [20–22]. Inverse dynamics methods, commonly employed in robotics and human biomechanics research, utilize known motions as inputs to the model to determine the coordinates (e.g., joint angles) and forces (e.g., muscle forces and joint torques) that are consistent with external measurements (e.g., trajectories from a motion capture system and ground reaction forces) [23–29]. These methods prove valuable in analyzing muscle activation and force changes in the upper and lower extremities during specific motions while wearing an exoskeleton [30–33]. However, inverse dynamics approaches rely on

kinematic data to generate motion, failing to provide insights into the underlying neuromuscular control strategies responsible for movement patterns and limiting our understanding of movement biomechanics. Additionally, when applied to solving musculoskeletal models, inverse dynamics methods often assume time-independent human motions and employ static optimization (SO) techniques [34] to address muscle redundancy, overlooking factors such as force balance within the muscle-tendon unit and muscle-contraction dynamics. Moreover, reliance on measured foot-ground reaction forces for calculating muscle forces and joint torques imposes significant demands on measurement equipment like force plates [35], thus confining related studies to laboratory or simplified environments.

On the other hand, forward dynamics approaches offer an alternative by utilizing muscle feedback signals obtained from surface electromyography (EMG) [36–41] or neural controllers [21, 42] as inputs to the model. These approaches employ an optimal control framework to predict the motion (e.g., joint angles) and forces (e.g., muscle forces, joint torques, and ground reaction forces) of coupled musculoskeletal-exoskeletal systems. Recent studies have also addressed the modeling of musculoskeletal dynamics with physiological noise, unveiling the influence of task constraints, sensory noise, and intrinsic muscle properties on muscle coordination and movement patterns [43]. Moreover, researchers have developed various dynamic optimization methods to minimize the metabolic cost of the human body in simulations and reduce the errors associated with kinematic data or constraints [44, 45]. However, it is worth noting that numerical algorithms used in forward dynamics analysis (e.g., shooting and collocation methods) are often computationally less efficient compared to inverse dynamics. As a result, these methods often compute only one gait cycle to represent human motion [45, 46]. However, in real-life scenarios, human gait needs to adapt to environmental changes, and sacrificing the number of gait cycles for computational efficiency may not be reasonable for simulating the motion of a coupled human-exoskeleton system in outdoor environments. To tackle this challenge, the covariance matrix adaptive evolution strategy (CMA-ES) has emerged as a promising approach [47]. CMA-ES utilizes its excellent parallel computing capability to evaluate multiple solutions simultaneously and accelerate the optimization process, making it suitable for solving complex and computationally demanding problems. In recent years, researchers have successfully applied the CMA-ES algorithm to solve complex nonlinear optimization problems of musculoskeletal forward dynamics models [48, 49]. While optimization-based forward dynamic frameworks have contributed significantly to our understanding of human movement dynamics, the exploration of reflex-based neuromuscular control strategies, especially in the context of exoskeleton interaction, has been less emphasized. How these mechanisms respond to external assistance exerted by an exoskeleton has rarely been thoroughly studied.

Furthermore, to improve the accuracy and computational efficiency of the models, researchers have developed hybrid frameworks that combine the merits of forward and inverse dynamics methods [46, 50]. One approach is the force-dependent kinematics (FDK) method, implemented in AnyBody software, which integrates inverse dynamics and quasi-static analysis to calculate joint torques but does not account for the muscle-tendon equilibrium equations [51, 52]. Another widely used hybrid method is the computed muscle control (CMC) [53, 54], which improves the accuracy of the static optimization (SO) model by incorporating a PID controller that feeds the forward dynamics model's error back to the SO model. Nonetheless, CMC sometimes exhibits poor convergence [16]. The forward-muscular inverse-skeletal (FMIS) framework is also an effective hybrid dynamic simulation method [50]. It generates muscle activation and muscle contraction dynamics in a forward manner while solving skeletal dynamics in an inverse manner. The FMIS framework [55] introduces torque-tracking errors into the objective function to relax dynamic equilibrium

constraints, thereby enhancing motion and torque tracking while improving computational efficiency. However, the torque tracking mode in FMIS does not align with the motion mechanisms generated by human neuromusculoskeletal control. It is important to note that although hybrid methods combining forward and inverse dynamics offer advantages, the reliance on kinematic data remains a limitation for estimating the dynamics of the human-exoskeleton coupled system outside laboratory settings. More critically, these methods may not fully respect the integrity of the human neuromusculoskeletal system's feedback control loop, potentially oversimplifying the complex interplay between reflexive responses and external mechanical assistance.

The dynamics methods previously explored have provided essential insights into the biomechanics of human-exoskeleton systems, and some studies have gotten involved in the domain of reflex-based neuromuscular mechanisms, but few have thoroughly investigated how these mechanisms respond to the assistance or resistance exerted by exoskeletons. Acknowledging these limitations, our study pivots towards a more focused analysis, integrating exoskeleton models with reflex-based gait models to effectively predict and understand the dynamic interplay between human movement and exoskeletal assistance, and thereby explain the qualitative general principles by which reflex-based neuromuscular systems react to external exoskeletons interventions. By examining outcomes such as muscle activation, muscle force, joint torque, and ground reaction forces, we aim to improve our understanding of the kinematics and biomechanics of human movement enhanced by exoskeleton technology, thereby enhancing the design and control of exoskeleton technology.

In this investigation, we focus on integrating exoskeleton models into a reflex-based gait model to explore the dynamic responses of subjects to various assistance torques provided by exoskeletons. The framework contributes to the development of human-exoskeleton interaction and muscle reflex model refinement. Specifically, the proposed simulation framework adopts a multi-degree-of-freedom multi-body system to model the coupled human-exoskeleton system. Comprehensively, it encompasses essential components including human lower limb muscle reflex mechanisms, muscle activation dynamics, muscle-tendon contraction dynamics, musculoskeletal dynamics, human body-exoskeleton interaction, and system-environment contact. The dynamic model is solved through an optimization-based process that introduces a multi-objective fitness function, which takes into account key gait criteria such as metabolic cost, walking stability, and joint hyperextension. The free muscle reflex parameters are then optimized using a modified CMA-ES with reference to neurophysiology. Our framework, which incorporates the complete neuromusculoskeletal feedback loop, allows for a comprehensive assessment of biomechanical and kinematic signals, including muscle activations, muscle forces, and joint torques, etc. These quantities, which are often challenging to measure experimentally, plays a vital role in advancing our understanding of human biomechanics and evaluating the efficacy of exoskeletons in facilitating human locomotion [56]. To validate the proposed simulation framework, comparisons are conducted between the simulation results and the experimental data obtained from a healthy subject wearing an exoskeleton while walking at different speeds and terrains. Although a quantitative assessment of the subject's behavior and biomechanics is neither possible nor reasonable due to challenges in precisely measuring physical parameters such as segment masses and moments of inertia, as well as the uniqueness of walking habits, the results indicate that the simulation framework effectively captures the qualitative aspects of the coupled system's kinematic and biomechanical signals. The results highlight the effectiveness and feasibility of our framework in capturing qualitative differences in muscle activity related to different functions, as well as the evolutionary patterns of muscle activity and kinematic signals across different walking speeds and terrains. Further walking simulations

of the human-exoskeleton system, both with and without exoskeleton assistance, contribute to an enhanced understanding of how the patterns of human muscle activation changes in response to exoskeleton assistance. In summary, our analysis highlights the adaptive neuromuscular adjustments during assisted/unassisted locomotion, revealing distinct muscle activity patterns for different functions and their adaptation under varying walking conditions. These findings underline the significance of our approach in capturing the intricate ways in which the human body adapts to external mechanical assistance, informing the development of more compliant exoskeleton systems.

2 Methods

2.1 Overview of the framework

Fig. 1(A) illustrates the working principle of the forward dynamics simulation framework for coupled neuromusculoskeletal-exoskeletal systems used in the following case studies. The framework consists of two subsystems: the muscle dynamics subsystem, encompassing the muscle reflex mechanism, muscle activation dynamics, musculotendon contraction dynamics, and musculoskeletal kinematics, and the rigid-body dynamics subsystem, representing the multi-rigid-body dynamics of the human-exoskeleton system. The rigid-body dynamics subsystem also incorporates the interaction between the human and exoskeleton, as well as the reaction forces exerted by the ground on the coupled system. The simulation framework can determine internal signals, including muscle activation, muscle forces, and joint torques. Rather than relying on empirical pre-determination or direct optimization methods, these signals are derived from a physiological perspective. This is achieved by optimizing the muscle reflex parameters, taking into account the intricate interplay among muscle activities, human-exoskeleton interaction, and the dynamic interaction between the system and the environment. Such an approach ensures a more accurate representation of the physiological behavior within the simulation framework. Muscle excitation, referring to the efferent signal transmitted by the motor neurons of the central nervous system, is generated through the muscle reflex mechanism. This mechanism takes into account various kinematic and physiological factors such as human pose, gait phases, musculotendon lengths, and muscle forces. The muscle excitation is then transformed into muscle activation through the process of muscle activation dynamics. Simultaneously, utilizing the information on human-exoskeleton movement and muscle position, the length and velocity of the musculotendon, as well as the moment arm of the force relative to the rotation axis of the corresponding joint, can both be calculated through musculoskeletal kinematics. Subsequently, the muscle activation, along with musculotendon length and velocity, drives the muscle to contract by employing the musculotendon contraction dynamics, which is simulated by the Hill-type muscle-tendon model [44].

By multiplying musculotendon contraction force with the moment arm of the muscle at each time instant, human joint torques can be derived. Furthermore, the ground reaction force and the interaction force between the human body and the exoskeleton can be customized based on user-defined models. Similarly, the assistive torque of the exoskeleton can be tailored according to the given exoskeleton control strategy and the current gait state. The ability to customize the human-exoskeleton interaction and foot-ground interaction is a notable advantage of the proposed simulation framework.

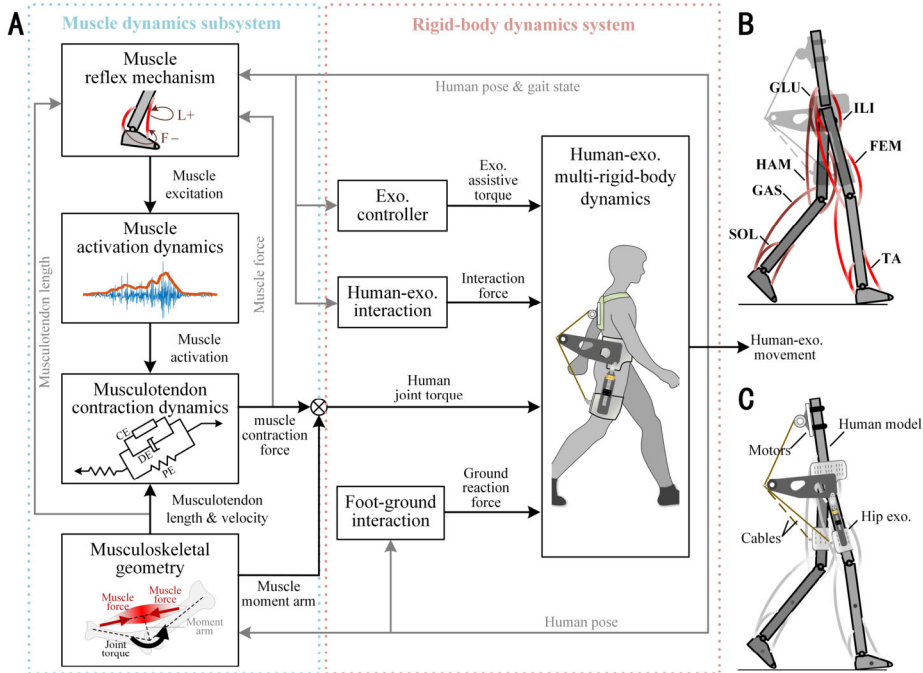


Fig. 1 Overview of the forward dynamics simulation framework for coupled neuromusculoskeletal-exoskeletal systems. (A) Flow diagram of the framework, demonstrating the working principle, where the grey arrows indicate the feedback. (B) Illustration of leg muscles, where the brown and red curves indicate the muscles of the left and right leg, respectively (GLU: gluteus maximus; ILI: iliopsoas; FEM: rectus femoris; HAM: hamstrings; GAS: gastrocnemius; SOL: soleus; TA: tibialis anterior). (C) Illustration of human-exoskeleton coupling, exemplified by a cable-driven hip exoskeleton (Color figure online)

The calculated human joint torques, human-exoskeleton interaction forces, ground reaction forces, and exoskeleton assistive torques are then employed as inputs to the human-exoskeleton multi-rigid-body model. This integration facilitates the movement of the coupled human-exoskeleton model, enabling a comprehensive analysis of the dynamic behavior.

To capture the influence of the key lower limb muscles involved in human walking in the sagittal plane at the neuromusculoskeletal level [57], the framework incorporates seven muscles in each leg, as depicted in Fig. 1(B) for reference. These muscles are crucial for controlling various movements. Specifically, five uniarticular muscles are identified: the iliopsoas (ILI) controls hip flexion, the gluteus maximus (GLU) controls hip extension, the rectus femoris (FEM) controls knee extension, the soleus (SOL) controls plantarflexion, and the tibialis anterior (TA) controls dorsiflexion. Additionally, two biarticular muscles, namely the hamstrings (HAM) and gastrocnemius (GAS), are also included in the framework; the hamstrings play a role in both hip extension and knee flexion, while the gastrocnemius contributes to knee flexion and plantarflexion movements.

The presented forward dynamics simulation framework can also include the exoskeleton hardware for evaluation of human-exoskeleton interaction. For the following case studies in this paper, a cable-driven hip exoskeleton is incorporated to exemplify the capabilities and performance of the proposed simulation framework. As depicted in Fig. 1(C), the cable-driven assistive hip exoskeleton model features two portable motors positioned on the user's back. These motors generate assistive forces that are transmitted to the thigh brace via ca-

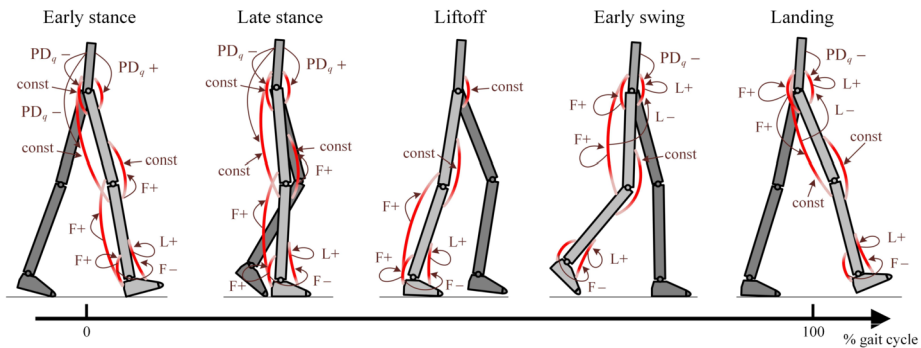


Fig. 2 Reflex mechanism of leg muscles during different gait phases, where the leg muscles along with the corresponding feedbacks and constant stimulations that influence muscle excitations are indicated (Color figure online)

bles. By considering the geometry of the coupled human-exoskeleton model, these forces can be effectively converted into assistive torques.

2.2 Modeling of muscle reflex and activation

2.2.1 Muscle reflex model

Physiologically, the activation of human skeletal muscles is governed by electrochemical signals known as muscle excitations σ , which are transmitted from the central nervous system to muscle fibers. The muscle reflex model provides a mathematical representation of how the states of the legs and muscles influence muscle excitation during voluntary movements. The combination of activation patterns is also heavily based on abundant experimental observations and identification [21, 58]. By incorporating the principles of legged mechanics and refining the segmentation of the gait, we develop an enhanced model of voluntary muscle reflexes, building upon the studies conducted by Geyer and Herr [21] and Geijtenbeek et al. [59]. The improved model, illustrated in Fig. 2, provides a more detailed and comprehensive representation of the muscle reflex during locomotion. In this section, we will delve into the muscle reflex mechanism, providing a detailed explanation, and outlining the modifications we have introduced.

In our study, we divide a single gait cycle into five distinct phases: three stance phases (early stance, late stance, and liftoff) and two swing phases (early swing and landing). Conventionally, a complete gait cycle is defined as the duration between two consecutive heel strikes of one foot [57]. Transitions between these gait phases are determined by a comprehensive assessment of the ground force and foot position of both legs, building upon the approach proposed by Geijtenbeek et al. [59], and also described in S1 of the Supplementary Material.

To facilitate subsequent descriptions, we will first provide the definitions of the following symbols: σ denotes the muscle excitation, K_L , K_F are positive proportional coefficients of muscle length and muscle force, $\tilde{l}$, $\tilde{l}_0$, and $\tilde{F}$ represent the normalized muscle length, normalized reference muscle length, and normalized muscle force, respectively. In the reflex model, the normalization is usually performed with respect to the optimal muscle length and optimal muscle force, i.e., $\tilde{l} = l/l_{\text{opt}}$, $\tilde{F} = F/F_{\text{opt}}$. The operator $\{\}_+$ is defined as follows: it takes a positive value when the variable is positive, and 0 when the variable is non-positive.

In the subsequent discussion, the superscript on the right indicate the specific muscle of interest.

The reflex mechanism of the TA muscle during a gait cycle is defined by

$$\sigma^{\text{TA}} = \left\{ K_L^{\text{TA}} \left(\tilde{l}^{\text{TA}} - \tilde{l}_0^{\text{TA}} \right) - K_F^{\text{TA}} \tilde{F}^{\text{SOL}} \right\}_+ . \quad (1)$$

In Eq. (1), the first term $K_L^{\text{TA}} \left(\tilde{l}^{\text{TA}} - \tilde{l}_0^{\text{TA}} \right)$ represents a positive muscle length feedback (L+) from TA to σ^{TA} , indicating that an increase in the length of TA can enhance its excitation; while the second term $-K_F^{\text{TA}} \tilde{F}^{\text{SOL}}$ denotes a negative muscle force feedback (F-), indicating that the contraction of SOL can inhibit the excitation of TA and providing a mechanism through which SOL activity can modulate the excitation of TA. The L+ mechanism of TA ensures foot clearance during the swing phases, and the F- mechanism demonstrates the antagonistic relationship between TA and SOL.

The reflex mechanism of the SOL muscle is given by

$$\sigma^{\text{SOL}} = \begin{cases} \left\{ K_F^{\text{SOL}} \tilde{F}^{\text{SOL}} \right\}_+ , & \text{earlystance/latestance/liftoff,} \\ 0, & \text{earlyswing/landing,} \end{cases} \quad (2)$$

and the mechanism of the GAS muscle exhibits similarities to the SOL mechanism

$$\sigma^{\text{GAS}} = \begin{cases} \left\{ K_F^{\text{GAS}} \tilde{F}^{\text{GAS}} \right\}_+ , & \text{earlystance/latestance/liftoff,} \\ 0, & \text{earlyswing/landing.} \end{cases} \quad (3)$$

Equation (2) and (3) indicate that both SOL and GAS are activated by positive force feedback (F+) from their own contraction during the stance phases. This activation makes SOL and GAS the main contributors to foot propulsion. However, during the swing phases of the corresponding leg, these muscles remain inactive.

The reflex mechanism of the FEM muscle is defined by

$$\sigma^{\text{FEM}} = \begin{cases} \left\{ C_1^{\text{FEM}} + \xi K_{F_1}^{\text{FEM}} \tilde{F}^{\text{FEM}} \right\}_+ , & \text{earlystance,} \\ \left\{ C_1^{\text{FEM}} + \xi K_{F_2}^{\text{FEM}} \tilde{F}^{\text{FEM}} \right\}_+ , & \text{latestance,} \\ \left\{ C_2^{\text{FEM}} \right\}_+ , & \text{liftoff/earlyswing/landing,} \end{cases} \quad (4)$$

where C_1^{FEM} and C_2^{FEM} are constant stimulations to the muscle (const); $K_{F_1}^{\text{FEM}} \tilde{F}^{\text{FEM}}$ and $K_{F_2}^{\text{FEM}} \tilde{F}^{\text{FEM}}$ represent positive force feedbacks (F+) that induce eccentric contraction of FEM, thereby helping to maintain knee stiffness during the early stance and late stance phases. The symbol ξ , which is equal to 0 or 1, represents a conditional switch, defined by

$$\xi = \begin{cases} 1, & q^{\text{knee}} < \hat{q}^{\text{knee}} \text{ or } \dot{q}^{\text{knee}} \leq 0, \\ 0, & \text{else,} \end{cases} \quad (5)$$

where q^{knee} , $\dot{q}^{\text{knee}}$, and $\hat{q}^{\text{knee}}$ are the angle, angular velocity, and angle threshold of the knee joint, respectively, with the direction of knee flexion as positive. When the knee is flexing above the threshold, the conditional switch ξ is set to 0, deactivating the F+ feedback and suppressing the muscle force of FEM. This mechanism ensures prompt inhibition of FEM, preventing hyperextension of the knee joint. After the late stance phase, the muscle is stimulated at a constant value C_2^{FEM} (const).

The reflex mechanism of the HAM muscle is characterized by

$$\sigma^{\text{HAM}} = \begin{cases} \{C^{\text{HAM}} - K_q^{\text{HAM}}(q^{\text{UB}} - q_0^{\text{UB}}) - K_{\dot{q}}^{\text{HAM}}\dot{q}^{\text{UB}}\}_+, & \text{earlystance/latestance,} \\ 0, & \text{liftoff/earlyswing,} \\ \{K_F^{\text{HAM}}\tilde{F}^{\text{GLU}}\}_+, & \text{landing,} \end{cases} \quad (6)$$

where q^{UB} , $\dot{q}^{\text{UB}}$, and q_0^{UB} are the angle, angular velocity, and reference angle of the upper body, respectively. $-K_q^{\text{HAM}}(q^{\text{UB}} - q_0^{\text{UB}}) - K_{\dot{q}}^{\text{HAM}}\dot{q}^{\text{UB}}$ represents a negative proportional-derivative (PD) feedback concerning the tilt of the upper body (PD_q) during the early stance and late stance phases, which plays a crucial role in ensuring the stability of the upper body. Furthermore, the contraction of GLU can activate HAM during the landing phase through positive force feedback $K_F^{\text{HAM}}\tilde{F}^{\text{GLU}}$ (F+), indicating a synergy relationship between the two muscles. HAM is additionally stimulated at a constant value C^{HAM} during the early stance and late stance phases.

The reflex mechanism of the GLU muscle is defined by

$$\sigma^{\text{GLU}} = \begin{cases} \{C^{\text{GLU}} - K_q^{\text{GLU}}(q^{\text{UB}} - q_0^{\text{UB}}) - K_{\dot{q}}^{\text{GLU}}\dot{q}^{\text{UB}}\}_+, & \text{earlystance/latestance,} \\ 0, & \text{liftoff,} \\ \{K_F^{\text{GLU}}\tilde{F}^{\text{GLU}}\}_+, & \text{earlyswing/landing.} \end{cases} \quad (7)$$

GLU is excited by a constant stimulation C^{GLU} (const), along with negative PD feedback $-K_q^{\text{GLU}}(q^{\text{UB}} - q_0^{\text{UB}}) - K_{\dot{q}}^{\text{GLU}}\dot{q}^{\text{UB}}$ (PD_q) during the early stance and late stance phases. The PD feedback is responsible for maintaining the global orientation of the upper body. During the swing phases, a positive force feedback $K_F^{\text{GLU}}\tilde{F}^{\text{GLU}}$ (F+) from itself is applied to GLU, decelerating the thigh.

The reflex mechanism of the ILI muscle is given by

$$\sigma^{\text{ILI}} = \begin{cases} \{C_1^{\text{ILI}} + K_{q_1}^{\text{ILI}}(q^{\text{UB}} - q_0^{\text{UB}}) + K_{\dot{q}_1}^{\text{ILI}}\dot{q}^{\text{UB}}\}_+, & \text{earlystance/latestance,} \\ C_2^{\text{ILI}}, & \text{liftoff,} \\ \left\{ \begin{aligned} &K_{L_1}^{\text{ILI}}(\tilde{l}^{\text{ILI}} - \tilde{l}_0^{\text{ILI}}) - K_{q_2}^{\text{ILI}}(q^{\text{UB}} - q_1^{\text{UB}}) \\ &- K_{\dot{q}_2}^{\text{ILI}}\dot{q}^{\text{UB}} - K_{L_2}^{\text{ILI}}(\tilde{l}^{\text{HAM}} - \tilde{l}_0^{\text{HAM}}) \end{aligned} \right\}_+, & \text{earlyswing/landing.} \end{cases} \quad (8)$$

In the early stance and late stance phases, the excitation of ILI is regulated by a positive PD feedback $K_{q_1}^{\text{ILI}}(q^{\text{UB}} - q_0^{\text{UB}}) + K_{\dot{q}_1}^{\text{ILI}}\dot{q}^{\text{UB}}$ (PD_{q_1}), along with constant stimulation C_1^{ILI} (const). Similar to HAM and GLU, the PD feedback for ILI also plays a role in maintaining the upper body orientation. During liftoff, a constant stimulation C_2^{ILI} (const) is applied to enhance the hip flexion torque that elevates the leg forward. During the swing phases, ILI is activated by a combined mechanism, involving negative PD feedback $-K_{q_2}^{\text{ILI}}(q^{\text{UB}} - q_1^{\text{UB}}) - K_{\dot{q}_2}^{\text{ILI}}\dot{q}^{\text{UB}}$ (PD_{q_2}), positive length feedback from itself $K_{L_1}^{\text{ILI}}(\tilde{l}^{\text{ILI}} - \tilde{l}_0^{\text{ILI}})$ (L+), and negative length feedback from HAM $-K_{L_2}^{\text{ILI}}(\tilde{l}^{\text{HAM}} - \tilde{l}_0^{\text{HAM}})$ (L-).

By implementing a refined segmentation of gait phases, our model effectively captures the distinct behavior of key muscles during specific phases of the gait cycle. Specifically, we distinguish the behavior of the FEM muscle in early stance and late stance (Eq. (4)), the HAM muscle in early swing and landing (Eq. (6)), and the GLU muscle in liftoff (Eq. (7)). This refinement, in comparison to Geijtenbeek et al. [59], enhances the representation accuracy of muscle reflexes and improves the fidelity of muscle excitation in dynamic simulations.

The muscle reflex models developed above provide a mathematical description and interpretation of the reflex mechanism of each muscle. There are 29 parameters to be determined by optimization to capture the characteristics of human motion, and they constitute the reflex parameter vector $\mathbf{w} \in \mathbb{R}^{29}$. In detail, $\mathbf{w}$ is comprised of proportional coefficients ($K_L^{\text{TA}}, K_F^{\text{TA}}, K_F^{\text{SOL}}, K_F^{\text{GAS}}, K_{F_1}^{\text{FEM}}, K_{F_2}^{\text{FEM}}, K_q^{\text{HAM}}, K_{\dot{q}}^{\text{HAM}}, K_F^{\text{HAM}}, K_q^{\text{GLU}}, K_{\dot{q}}^{\text{GLU}}, K_F^{\text{GLU}}, K_{q_1}^{\text{ILI}}, K_{\dot{q}_1}^{\text{ILI}}, K_{L_1}^{\text{ILI}}, K_{q_2}^{\text{ILI}}, K_{\dot{q}_2}^{\text{ILI}}, K_{L_2}^{\text{ILI}}$), normalized reference muscle lengths ($\tilde{l}_0^{\text{TA}}, \tilde{l}_0^{\text{ILI}}, \tilde{l}_0^{\text{HAM}}$), joint angle thresholds ($q_1^{\text{UB}}, \hat{q}_0^{\text{knee}}$), and constant stimulations ($C_1^{\text{FEM}}, C_2^{\text{FEM}}, C^{\text{HAM}}, C^{\text{GLU}}, C_1^{\text{ILI}}, C_2^{\text{ILI}}$). The detailed optimization procedures will be introduced in Sect. 3.1.

2.2.2 Muscle activation model

The level of muscle activation, denoted as a , depicts the concentration of bound Ca^{2+} -ions in the muscle sarcoplasm relative to its physiological maximum, and typically ranges from $a_{\min}$ to 1 [60]. However, it is important to note that the activation and deactivation of the muscle do not occur instantaneously following muscle excitation. The detailed modeling of the muscle activation process is provided in S2 of the Supplementary Material.

2.3 Modeling of musculotendon contraction dynamics

To accurately depict the process of converting muscle excitation into muscle force, several factors such as muscle activation, muscle fiber length and velocity, tendon length and velocity, and others need to be considered [61, 62]. In this study, we employ the modified Hill-type model developed by Millard et al [63] to describe musculotendon contraction dynamics, which is introduced in detail in S3 of the Supplementary Material.

2.4 Musculoskeletal modeling

2.4.1 Muscle moment derivation

In a forward-dynamic gait simulation, the motion of the human skeletal model is controlled by torques produced by multiple muscles spanning the joints [16]. The magnitude of torque generated at a joint is influenced by several factors, including the joint angle, the contraction level and the number of involved muscles, the attachment positions of the muscles, and their paths of action [64]. This subsection focuses on examining the influence of musculoskeletal geometry on muscle arms, which is crucial for accurately converting muscle forces into moments.

We utilized the musculoskeletal geometry provided by the SCONE H0914M model [65], which allows us to calculate the muscle arms around each joint. For the simplicity of computation, muscles are assumed as piecewise linear curves that approximates their actual wrapping paths. The torque generated by muscle i around the joint can be written as

$$\tau_i = F_i^{\text{M}} \cos \alpha \cdot d_{\text{mus}}, \quad (9)$$

where $F_i^{\text{M}} \cos \alpha$ represents the component of force produced by muscle i in the direction of the tendon, and d_{mus} is the moment arm of the muscle with respect to the corresponding joint. The derivation of d_{mus} is provided in S4 of the Supplementary Material. Then the joint torque can be calculated by summing up all the torques produced by related muscles.

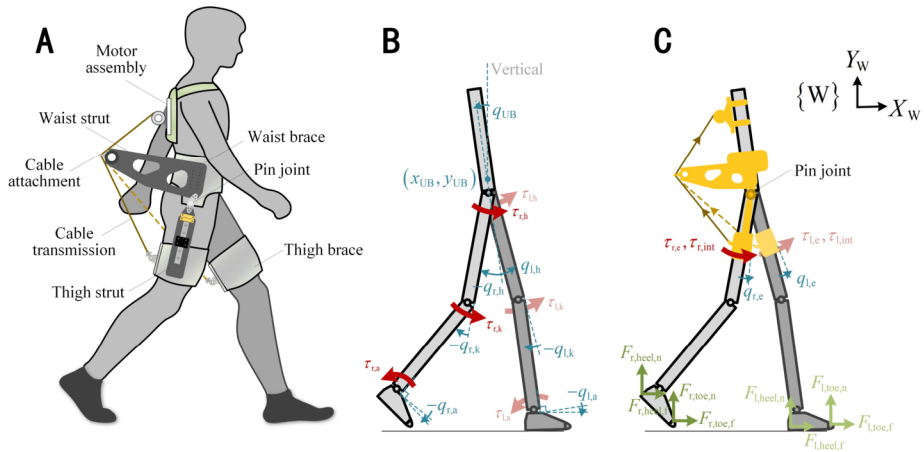


Fig. 3 Illustration of the coupled human-exoskeleton system. (A) A human wearing the powered cable-driven hip exoskeleton. (B) Generalized coordinates and joint torques of the human. (C) Generalized coordinates of the exoskeleton, interactive torques between the human and the exoskeleton, and the ground reaction forces (Color figure online)

2.4.2 Multi-rigid-body dynamics of coupled human-exoskeleton systems

Based on the modeling of muscle reflexes and activation, muscle-tendon contraction dynamics, and musculoskeletal models, we develop a multi-rigid-body dynamic model of the coupled human-exoskeleton system in this subsection. Specifically, we illustrate the modeling process with a human body wearing a powered cable-driven hip exoskeleton, which could provide assistive force during hip extension. Fig. 3(A) depicts the coupled human-exoskeleton system, in which the exoskeleton is coupled with the human body through waist and thigh braces. On each side of the body, a motor assembly is mounted on the exterior of the backpack, driving a cable that extends downward and connects to the thigh brace. A cable attachment, located on the waist strut, allows the cable to pass through it and functions as a fixed pulley as well as a motion-limiting mechanism [66]. The waist and thigh struts are bolted to the waist brace and the thigh brace, respectively, and are interconnected by a pin joint assembly. For the sake of modeling and analysis, the following assumptions are made:

- The relative displacements between the waist brace and the human trunk as well as the backpack and the human trunk are neglected because their interaction forces are relatively weak and do not significantly affect the response of the dynamical system. Therefore, the waist brace, the backpack, and the human trunk can be considered as a single rigid body.
- The motion, forces, and torques in the coronal and transverse planes are neglected due to their small range of motion [55].
- The axis of rotation of the pin joint in the exoskeleton and the axis of the human hip joint are ideally co-linear, thus preventing any joint misalignment issues in the model.

Based on the above assumptions, a dynamic model of the coupled human-exoskeleton system is developed (Fig. 3(B) and (C)), which consists of 9 rigid segments and is described by 11 DOFs. The joint torques are denoted as τ_{i_1, i_2} , where the subscript $i_1 \in \{r, l\}$ represents right (r) and left (l), and $i_2 \in \{h, k, a\}$ represents hip (h), knee (k), and ankle (a). The exoskeleton produces assistive torques $\tau_{i_1, e}$ for the wearer, creating interaction torques $\tau_{i_1, int}$ with the thighs at the contact interface (the detailed form will be provided later). The angles

between adjacent two rigid elements are denoted by q_{i_1, i_2} , and the angle between the upper body and the vertical plane is denoted by q_{UB} . Exceptionally, $q_{i_1, a}$ is defined as the angle between the foot and the line perpendicular to the shank, which aligns with the general definition of ankle plantarflexion/dorsiflexion [67]. Following the convention, positive torques and angles are defined as those that produce counter-clockwise rotation. The position vector of the upper body origin with respect to the world coordinate system $\{W\}$ is (x_{UB}, y_{UB}) .

It is important to distinguish the center of mass (COM) and the geometric center of a segment. The COMs of the body segments, including the thigh, shank, and foot, are determined using experimental data collected by Anderson and Pandy [68]. On the other hand, the COM of the exoskeleton's thigh segment is obtained from SolidWorks (2021, Dassault Systèmes SolidWorks Corp., Waltham, MA, USA) using the mass property tool.

The ground reaction forces (Fig. 3(C)) are denoted by F_{i_1, i_3, i_4} , where the subscript $i_3 \in \{h, t\}$ represents heel (h) and toe (t), and $i_4 \in \{n, f\}$ represents normal force (n) and friction force (f). The detailed forms of the ground reaction forces will be discussed later.

Equation of motion The equation of motion (EOM) of the coupled human-exoskeleton system is derived by means of generalized Lagrange Equation of the second kind:

$$\mathbf{D}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{Q}, \quad (10)$$

where $\mathbf{q} = (x_{UB}, y_{UB}, q_{UB}, q_{r, h}, q_{r, k}, q_{r, a}, q_{r, e}, q_{l, h}, q_{l, k}, q_{l, a}, q_{l, e})^T$ is the generalized coordinate vector, $\mathbf{D}(\mathbf{q}) \in \mathbb{R}^{11 \times 11}$ represents the inertia matrix, $\mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{11 \times 11}$ denotes the matrix for the centrifugal force and Coriolis force, $\mathbf{G}(\mathbf{q}) \in \mathbb{R}^{11 \times 1}$ is the gravity vector, and $\mathbf{Q} \in \mathbb{R}^{11 \times 1}$ represents the generalized force vector, constructed by

$$\mathbf{Q} = \boldsymbol{\tau}_{\text{joint}} + \boldsymbol{\tau}_{\text{exo}} + \boldsymbol{\tau}_{\text{int}} + \mathbf{J}_{\text{grf}}^T \mathbf{F}_{\text{grf}}, \quad (11)$$

where $\boldsymbol{\tau}_{\text{joint}}$, $\boldsymbol{\tau}_{\text{exo}}$, and $\boldsymbol{\tau}_{\text{int}}$ denote the vectors of joint torques, assistive torques generated by the exoskeleton, and the interaction torques between the exoskeleton and human, respectively; $\mathbf{F}_{\text{grf}}$ represents the vector of ground reaction forces, and $\mathbf{J}_{\text{grf}}$ is the Jacobi matrix that transforms the ground reaction forces into the generalized forces. The detailed expressions of the EOM are provided in S5 of the Supplementary Material.

Interaction torques and ground reaction forces The interaction torque generated between the exoskeleton and the thigh is characterized by the Kelvin-Voigt model [69], which consists of a linear spring in parallel with a damper. The model describes the relationship between the applied torque and the resulting displacement or velocity:

$$\tau_{i_1, \text{int}} = k_{\text{int}} q_{i_1, e} + d_{\text{int}} \dot{q}_{i_1, e}, \quad (12)$$

where k_{int} and d_{int} represents the stiffness and damping coefficients, respectively [70]; $q_{i_1, e}$ denotes the angle between the exoskeleton's thigh segment and the human thigh.

The normal force, which represents ground reaction force in the direction perpendicular to the ground surface, is characterized by a contact model proposed by Hunt and Crossley [71]. This contact model captures the dynamic interaction between two bodies during a collision and is expressed as

$$F_{i_1, N} = k \delta_{i_1}^{3/2} (1 + 1.5c \dot{\delta}_{i_1}), \quad (13)$$

where k represents a stiffness constant incorporating both material and geometry properties, c is the dissipation coefficient, and δ_{i_1} represents the depth of contact between the heel/toe

and the ground (which remains zero when there is no contact). In our model, both the heel and the toe are modeled as elastic spheres with a radius of 3 cm.

The friction force exerted by the ground on the heel/toe can be expressed as [72]

$$F_{i_1, f} = F_{i_1, N} \left[\min \left(\frac{v_{i_1}}{v_t}, 1 \right) \left(\mu_d + \frac{2(\mu_s - \mu_d)}{1 + (v_{i_1}/v_t)^2} \right) + \mu_v v_{i_1} \right], \quad (14)$$

where μ_s , μ_d , and μ_v represent the coefficients of static friction, dynamic friction, and viscous friction, respectively; v_t denotes the transition velocity, at which the friction reaches its maximum value; v_{i_1} is the tangential slip velocity between the heel/toe and the ground at the contact point.

So far, we have developed a dynamic model of the coupled human-exoskeleton system that integrates the biological joint torques, exoskeleton assistance, human-exoskeleton interaction, and foot-ground contact. This comprehensive framework allows for the simultaneous consideration of these key factors in the analysis of human-exoskeleton-ground interaction dynamics.

2.5 Model optimization

The muscle reflex mechanism described in Sect. 2.2.1 involves 29 free parameters (i.e., $\mathbf{w} \in \mathbb{R}^{29}$). These parameters enable customization of human motion to adapt to various walking speeds and environmental conditions. Specifically, the optimization problem is constructed as follows

$$\begin{aligned} \min J &= J(\mathbf{w}), J(\mathbf{w}) = \begin{cases} J_1, & \text{step1,} \\ J_2, & \text{step2,} \end{cases} \\ \text{s.t.} \quad & F_{\text{opt}} \left(a \mathbf{f}^L \left(\tilde{l}^M \right) \mathbf{f}^V \left(\tilde{v}^M \right) + \beta \tilde{v}^M + \mathbf{f}^{\text{PE}} \left(\tilde{l}^M \right) \right) \cos \alpha - F_{\text{opt}} \mathbf{f}^T \left(\tilde{l}^T \right) = 0, \\ & \mathbf{D}(\mathbf{q}) \ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}}) \dot{\mathbf{q}} + \mathbf{G}(\mathbf{q}) = \mathbf{Q}, \\ & 0 \leq \sigma \leq 1, \\ & a_{\min} \leq a \leq 1. \end{aligned} \quad (15)$$

The muscle reflex parameters $\mathbf{w}$ are determined through a two-step optimization process. In step 1, our objective is to maximize the walking distance of the model without falling down; in step 2, we aim to constrain the model's walking behavior utilizing multi-dimensional evaluation criteria so that it satisfies the speed constraint and is more natural and human-like. Throughout both steps, it is essential to satisfy the muscle-tendon force equilibrium constraint, the rigid-body dynamic constraint, as well as the boundaries of muscle excitation and activation. This approach will be demonstrated to be efficient and robust, eliminating the need for accurate initial guesses.

Step 1. Since the initial guess of the parameters may deviate significantly from their optimum values, the primary objective of this step is to maximize the walking distance achieved by the model within a single simulation. The evaluation of this objective is performed using J_1

$$J_1 = \psi - \sqrt{v^2 + 1} \cdot x_{\text{COM}}(T), \quad (16)$$

where x_{COM} represents the x -coordinate of the model's COM, T denotes the number of timesteps before either the simulation ends or the model falls down, v denotes the tangent value of the ground slope, and ψ is a prescribed parameter indicating the desired distance in meters from the model to the destination.

The optimization switches from step 1 to step 2 when $J_1 < 0$, indicating that the model can walk more than ψ meters in T timesteps. The population corresponding to this result is then retained and used as the initial guess for optimization in step 2.

Step 2. In this step, the optimization objective is to minimize the following hybrid objective function:

$$J_2 = \omega_{\text{vel}} J_{\text{vel}} + \omega_{\text{angle}} J_{\text{angle}} + \omega_{\text{GRF}} J_{\text{GRF}} + \omega_{\text{effort}} J_{\text{effort}}, \quad (17)$$

where ω_{vel} , ω_{angle} , ω_{GRF} , and ω_{effort} are the parameters that weight the four objectives (J_{vel} , J_{angle} , J_{GRF} , and J_{effort}), respectively. The four objective functions are interpreted as follows:

- J_{vel} represents the velocity fitness, which evaluates the model's ability to achieve the desired velocity at the model's COM ($v_{\text{COM}}^{\text{des}}$), with a significant penalty for falling down:

$$J_{\text{vel}} = \frac{1}{T} \sum_t (v_{\text{COM}}(t) - v_{\text{COM}}^{\text{des}})^2 + \delta_{\text{vel}}, \quad (18)$$

where $v_{\text{COM}}(t)$ denotes the velocity of the model's COM at the t -th timestep. The first term penalizes the model when its velocity deviates from $v_{\text{COM}}^{\text{des}}$. $\delta_{\text{vel}} = 1$ when the model falls down during the simulation, and $\delta_{\text{vel}} = 0$ otherwise.

- J_{angle} is the function that constrains ankle hyperflexion and hyperextension

$$J_{\text{angle}} = \frac{1}{T} \sum_t \sum_{i_1=r,l} [\max(q_{i_1,a}(t) - \hat{q}_{a,\max}, 0) - \min(q_{i_1,a}(t) - \hat{q}_{a,\min}, 0)], \quad (19)$$

where $\hat{q}_{a,\max}$ and $\hat{q}_{a,\min}$ are the upper and lower threshold of the ankle motion, respectively. This term equals zero when the ankle motions range within $[\hat{q}_{a,\min}, \hat{q}_{a,\max}]$.

- J_{GRF} imposes a penalty when the normal ground reaction forces exerted on the heels and toes exceed the threshold value $\hat{F}_{\text{GRF}}$:

$$J_{\text{GRF}} = \frac{1}{T} \sum_t \frac{1}{mg} \sum_{i_1=r,l} \max(F_{i_1,\text{toe},n}(t) + F_{i_1,\text{heel},n}(t) - \hat{F}_{\text{GRF}}, 0), \quad (20)$$

where m denotes the mass of the model, and g is the gravitational acceleration. This term equals zero when the threshold value is not exceeded.

- J_{effort} represents the metabolic expenditure of the muscles, following the model developed by Wang et al [73]:

$$J_{\text{effort}} = \frac{1}{x_{\text{COM}}(T)} \sum_t \frac{\Delta t_{\text{sim}}}{m} \sum_i (r_A^i(t) + r_M^i(t) + r_S^i(t) + r_W^i(t)), \quad (21)$$

where $x_{\text{COM}}(T)$ denotes the model's walking distance in T timesteps; Δt_{sim} denotes the simulation time step; r_A , r_M , r_S , and r_W denote the rates for muscle activation heat, muscle maintenance heat, muscle shortening heat, and positive mechanical work, respectively (the superscript ' i ' takes TA, SOL, GAS, FEM, HAM, GLU, or ILI).

The selection of ω_{vel} , ω_{angle} , ω_{GRF} , and ω_{effort} is based on our primary goal for the model to complete the specified distance. Secondary to this, we aim for the ground contact forces to

be within a reasonable range, and finally, we optimize for joint angle limits and energy consumption. Therefore, the weights are empirically assigned in descending order of priority, with $\omega_{\text{vel}} = 100$, $\omega_{\text{angle}} = 0.1$, $\omega_{\text{GRF}} = 10$, and $\omega_{\text{effort}} = 0.1$.

To determine the parameters $\mathbf{w}$, the Covariance Matrix Adaptation Evolution Strategy (CMA-ES) [74] is adopted. The CMA-ES optimization is recognized for its effectiveness in solving complex optimization problems, making it a valuable tool in various domains. In the context of CMA-ES optimization, 13 populations are used per iteration, with an initial standard deviation of 0.01. Each population represents a specific set of free parameters $\mathbf{w}$. In each iteration of the optimization process, the performance of every population is assessed by the fitness function J , based on which, the optimization algorithm adjusts the populations to improve their performances in the subsequent iteration. This iterative process continues until a satisfactory solution is obtained or the maximum number of iterations is reached.

2.6 Simulation setup

Our model is designed to be independent of specific walking conditions and environments, enabling a systematic and comprehensive study of their effects. We aim to verify the feasibility of the dynamic simulation framework and to assess the extent to which the developed model can capture the dynamic responses of the coupled human-exoskeleton system in various scenarios. For all the case studies in this research, the proposed framework is implemented and tested in MATLAB (R2021b, The MathWorks Inc., Natick, MA, USA), with a simulation time step of $\Delta t_{\text{sim}} = 5$ ms. The dynamic model, described by ordinary differential equations, is numerically solved using the fourth-order Runge-Kutta method with a step size of 0.5 ms. The values of parameters involved in the framework and initial pose information of the human-exoskeleton model are provided in S6 and S7 of the Supplementary Material, respectively. The optimized muscle reflex parameters for each scenario are provided in S8 of the Supplementary Material. The simulation results are later validated by comparison with walking experiments under the same conditions.

2.7 Participant and experiment protocol

A healthy male subject (aged 28, weighing 70 kg, and standing at a height of 171 cm) participates in the experiment. The protocol received approval from the Ethical Committee of Fudan University. The subject is fully informed of the aims and details of the protocol and voluntarily agreed to participate, with the signing of written informed consent.

Fig. 4 (A) shows the photo of the experimental setup. As the primary objective of the experiment is to validate the effectiveness of the dynamic simulation framework, there is no requirement for the exoskeleton to provide assistance. Hence, the subject involved in the experiment does not carry the exoskeleton's motors and cables, i.e., the subject wears a passive hip exoskeleton that is tightly coupled to the human thigh. To measure the muscle activity, EMG electrodes (Ultium EMG, Noraxon, Scottsdale, AZ, USA) are positioned on five lower limb muscles on both sides: TA, GAS, SOL, FEM, and HAM. The electrodes are firmly secured using medical adhesive tapes to ensure close contact with the skin (Fig. 4(B)). Additionally, the subject wears an inertial motion capture suit (Perception Neuron, Noitom, Beijing, China) equipped with multiple inertial sensors. These sensors are attached to elastic straps that are fastened to the corresponding body segments (Fig. 4(B)).

The experimental protocol commences with a pose calibration procedure for the inertial motion capture system. Following the calibration, 2 minutes of still standing is conducted to record the resting surface electromyography (sEMG) signals of the muscles. Subsequently,

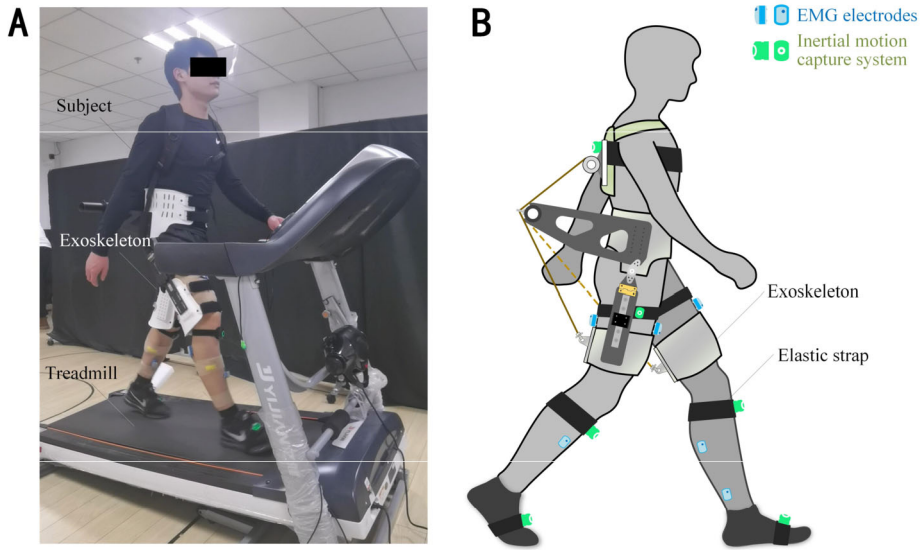


Fig. 4 Human walking experiments. (A) The subject wearing the exoskeleton and sensors walks on a treadmill with adjustable slope. (B) Illustration of the sensor placement (Color figure online)

the subject, wearing the exoskeleton, performs walking tasks on a treadmill under four distinct experimental scenarios. Each scenario has a duration of 1 minute, and a rest period of 30 seconds is provided between scenarios to mitigate muscle fatigue.

2.8 Experimental data acquisition and process

The sEMG signals of the 10 muscles are measured using the EMG system. The data are processed according to the methodology described in [75]. A sampling frequency of 2000 Hz is utilized for data acquisition. To preprocess the raw sEMG data, a second-order Butterworth high-pass filter with a cut-off frequency of 10 Hz is applied. Subsequently, the data are subjected to full-wave rectification and further filtered using a Butterworth low-pass filter with a cut-off frequency of 5 Hz. To standardize the preprocessed signals, normalization is applied to the sEMG data. The normalization process involves scaling the signals to the peak values observed during walking on the flat ground at a speed of 1.0 m/s. This normalization technique, which is also widely used in human biomechanics studies, ensures that all muscle activation signals are comparable and represents relative changes in muscle activity across different scenarios and walking conditions. The normalization also facilitates meaningful comparison and analysis of sEMG signals and muscle activation during various locomotion tasks in a qualitative manner.

For the measurement of joint angles and the tilt angle of the upper body, an inertial motion capture system is employed. This system provides quaternion measurements of the feet, shanks, thighs, and upper body, enabling the calculation of joint angles of hip, knee, and ankle joints, as well as the tilt angle of the upper body. The sampling frequency of the inertial motion capture system is set to 100 Hz.

3 Case study 1: walking on flat and uphill grounds

In the first case, we conduct simulations of the coupled neuromusculoskeletal-exoskeletal system walking on the flat and uphill grounds (5.7° slope, 10%) at the walking speed of 1.0 m/s. It is important to note that the primary focus of our simulations lies in verifying the feasibility of the proposed framework in predicting the dynamic responses of the coupled neuromusculoskeletal-exoskeletal system, rather than evaluating the performance of the employed exoskeleton in providing assistance. As a result, in this case, our simulations and experiments involve the coupling and interaction between the exoskeleton and the human, without necessarily requiring the exoskeleton to provide assistance.

3.1 Simulation analysis

Simulations are performed under two scenarios, comprising walking on the flat ground and walking uphill (5.7° slope, 10%) at $v_{COM}^{des} = 1.0$ m/s. Each scenario is simulated for a duration of 10 seconds using the 2-step optimization process outlined in the previous subsection. In the first step of optimization, we simulate walking on the flat and uphill grounds at a speed of 1.0 m/s to quickly converge to a feasible parameter set. Subsequently, for both ground slope scenarios, we conduct simulations with multiple speed constraints in the second step of optimization, using the outcomes from the first step to refine the parameters. The results reveal that the first step optimization for walking on the flat and uphill grounds requires 33 and 65 iterations, respectively, with an average computing speed of approximately 3 iterations per hour on a workstation (Intel Xeon Platinum 8124M, 3.0 GHz, 128 GB RAM). The second step of optimization requires approximately 19 iterations for each scenario. Statistical analyses are conducted using one-way Analysis of Variance (ANOVA) to assess the significant differences in peak muscle activations obtained across multiple gait cycles under various simulation conditions.

Fig. 5 illustrates the simulation results for walking on both flat ground and uphill at a speed of 1.0 m/s. The developed dynamic simulation framework not only facilitates stable walking that closely resembles human behavior (Fig. 5(A) and Supplementary Video S1) but also provides a comprehensive understanding of the coupled neuromusculoskeletal-exoskeletal system (Fig. 5(B-F)). The obtained results encompass various aspects, including muscle activations and forces (Fig. 5(B-C), exemplified by the HAM, GLU, and SOL muscles of the right leg), joint torques (Fig. 5(D), exemplified by joint torques of the right hip, knee, and ankle joints), human-exoskeleton interaction torque (Fig. 5(E), exemplified by the interaction torque of the right leg), kinematic data (Fig. 5(F), exemplified by the joint angles of the right hip, knee, and ankle), and ground reaction forces (normal forces and friction forces, Fig. 5(G)). This is particularly significant considering that certain biomechanical information, such as muscle activation, muscle forces, and joint torques, as well as interaction information, such as human-exoskeleton and foot-ground interaction forces, are challenging to measure through instrumentation in experimental settings. Furthermore, the patterns of muscle activation demonstrate qualitative concordance with those observed in the study by Song et al [38], suggesting a locomotion behavior akin to that of humans; while quantitative comparison is not conducted since the system in [38] does not include an exoskeleton. We notice a pronounced peak of the human-exoskeleton interaction torque at the landing phase in each gait cycle in Fig. 5(E), which displays a “noisy” profile. This is due to the sharp overextension of the swinging leg during the landing phase, which is also evident in natural human walking behavior. This phenomenon is also manifested in the abrupt variation of hip and knee angle simultaneously in Fig. 5(F).

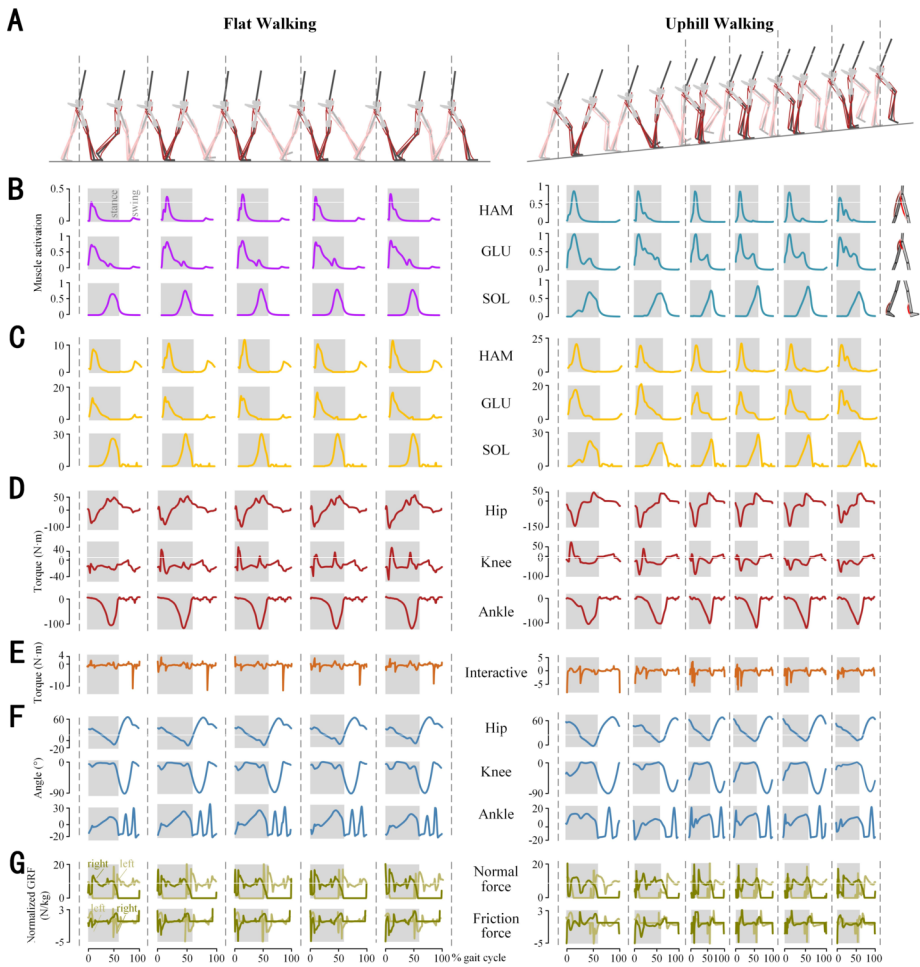


Fig. 5 Simulation results of the coupled human-exoskeleton system walking on the flat and uphill ground. (A) Snapshots of the model walking on the flat ground and uphill at a speed of 1 m/s. (B) Normalized muscle activation of HAM, GLU, and SOL of the right leg, where the shaded and unshaded areas denote the stance and swing phases, respectively. (C) Muscle forces of HAM, GLU, and SOL of the right leg, normalized by the subject's body weight. (D) Joint torques of the hip, knee, and ankle of the right leg. (E) Human-exoskeleton interaction torques on the right thigh. (F) Joint angles of the hip, knee, and ankle of the right leg. (G) Ground reaction forces, normalized by the subject's body weight (right and left legs in dark and light green colors, respectively) (Color figure online)

An important aspect to highlight is that the obtained results are achieved without relying on experimental data input, leading to substantial savings in experimental costs. As a result, the proposed dynamic model and framework offer a unique capability that opens up the potential for estimating the biomechanics and motion dynamics of humans or human-exoskeleton systems beyond the confines of a laboratory environment, allowing for a broader understanding of the system's behavior and performance in various settings.

The validity of our dynamic simulation framework is further demonstrated by its ability to capture and represent the performance of the human-exoskeleton system under flat and

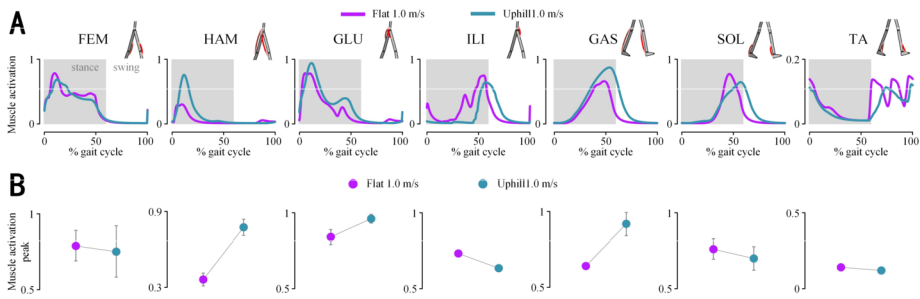


Fig. 6 Muscle activations in the right leg with respect to different terrains, the walking speed is 1.0 m/s. (A) From the left to the right, showing the muscle activations of the FEM, HAM, GLU, ILI, GAS, SOL, and TA muscles when walking on the flat and uphill (5.7° slope) grounds. The curves are averaged results of 7 walking cycles. (B) The muscle activation peaks (average $\pm$ std) of the respective muscles in flat and uphill walking (Color figure online)

uphill grounds. Significant changes in step length and muscle activation peaks are observed on different slopes, highlighting the adaptive nature of our model in adjusting gait according to the environment. Specifically, during uphill walking, the step length is significantly shortened from 0.58 m of flat walking to 0.44 m, indicating the challenges of walking uphill. Moreover, Fig. 6 presents the time series and peak values of muscle activations in the right leg (including the muscles FEM, HAM, GLU, ILI, GAS, SOL, and TA) during a gait cycle for walking on the flat ground in relation to different slopes (flat or uphill ground (5.7° slope, 10%)). It is important to note that different muscles exhibit different patterns of change in response to variations in walking conditions, which can be attributed to their specific roles within the gait cycle. Specifically, we observe a significant increase in the activation of GLU and HAM muscles ($p < 0.05$), suggesting the increased effort to maintain gait stability. There is also an increase in GAS muscle activation peaks ($p < 0.05$), reflecting the increased power generation for walking uphill. The lack of significant difference in peak activation of the FEM muscle, as observed in this study and similar to the findings of Saito et al [76], can be explained by the influence of step length during uphill walking. The shorter step length during uphill walking compared to flat walking is advantageous for maintaining stability and support during the stance phase, as it allows for a more controlled and balanced weight transfer. As a result, the demand for force generation from the FEM muscle may not significantly change during uphill walking, leading to a similar peak activation level compared to flat walking conditions. Conversely, the activation peak levels of the SOL, TA, and ILI muscles show significant decreases ($p < 0.05$) during uphill walking. This decline can be primarily attributed to the substantial decreases in step length, which affects the mechanical demands placed on these muscles. The shorter step length during uphill walking requires less force generation from the SOL, TA, and ILI muscles for propulsion and support. This decrease in demand is accompanied by compensatory behavior among various muscles. Notably, the SOL and GAS muscles, which share similar functions in plantar flexion, demonstrate compensatory behavior. During uphill walking, the activation level of the GAS muscle notably increases, leading to a corresponding decrease in activation level of the SOL muscle.

It is important to highlight again that our dynamic model and optimization framework do not rely on experimental data, offering us a significant advantage in terms of efficiently integrating and testing updated exoskeleton designs and control strategies in a wide range of scenarios. The results obtained from the framework provide valuable insights into muscle activation patterns and motor control mechanisms during walking across different speeds

and environmental conditions. This flexibility in testing and the wealth of information obtained from the framework are instrumental in verifying the adaptability of the system to different environments and different gait patterns. Ultimately, this leads to the provision of optimal assistance tailored to the specific needs of the user.

3.2 Experimental validation

To validate the effectiveness of the dynamic simulation framework and assess the accuracy of the predicted results, we further conduct two walking experiments that encompassed walking on the flat and uphill (5.7° slope, 10%) ground at 1.0 m/s, aligning with the scenarios used in the simulations. Comparisons are conducted between the simulation results and experimental data, as shown in Fig. 7. Moreover, the Pearson correlation coefficient R is calculated to determine the correlation between experimentally measured signals, such as EMG and joint kinematics angles, and the corresponding results obtained from simulations. The aim is to validate that the walking motions generated by the simulation framework exhibit similar kinematics and muscle activities to those observed in real experiments. Previous gait experiments conducted by Wren et al. [77] have revealed that the coefficients of correlation (R) of sEMG signals among different subjects range from $R = 0.40$ to 0.81 . In Fig. 7(A), it is evident that, with the exception of TA muscles (to be discussed in Sect. 6), the predicted muscle activation in the simulation aligns well with the measured sEMG signals in terms of the qualitative trends, with $R > 0.4749$. These findings provide substantial evidence supporting the effectiveness of the proposed framework in capturing the qualitative neuromuscular reflex patterns exhibited by an exoskeleton-wearing individual during walking.

Furthermore, Fig. 7(B) illustrates the effects of slope on peak muscle activities in both simulations and experiments. The patterns of change in peak muscle activity relative to the terrain are qualitatively consistent in both simulation and experimental results. Specifically, the FEM muscle does not exhibit a significant correlation between the peak muscle activity and slope, because it is mainly responsible for support. The GAS and HAM muscles receive significant increases in peak muscle activities when walking uphill to increase propulsion and enhance stability, respectively. On the contrary, there is a slight decrease in peak muscle activity of the TA muscles during uphill walking. Note that the peak values of sEMG of the SOL muscle increase significantly in experiments during uphill walking, while there is a slight decrease in simulations. This discrepancy may be attributed to the fact that the GAS and SOL muscles have different compensatory behavior in simulations and experiments.

Moreover, Fig. 7(C) demonstrates the variation of joint angles in simulations and experiments. Except for the knee joint angle during uphill walking, the simulated and experimental joint angles are qualitatively consistent in terms of their variation trends ($R > 0.4279$). The lack of correlation in knee joint angles during uphill walking may be due to the delay in the maximum knee flexion angle between simulations and experiments. The observed delay in knee flexion during uphill walking can be attributed to the optimization of the gait pattern based on energy optimization criteria. In order to adapt to the uphill terrain, the simulation framework optimizes the gait pattern by adjusting various muscle activations and joint angles. This optimization process prioritizes energy efficiency, which may lead to a delay in achieving the maximum knee flexion angle compared to the experimental results. Although incorporating reference kinematics into optimization objectives can enhance the fitting quality for a specific task, the introduction of experimental data can significantly limit the flexibility of the simulation framework. The reliance on experimental data would restrict the simulation framework's ability to explore alternative solutions or adapt to different scenarios.

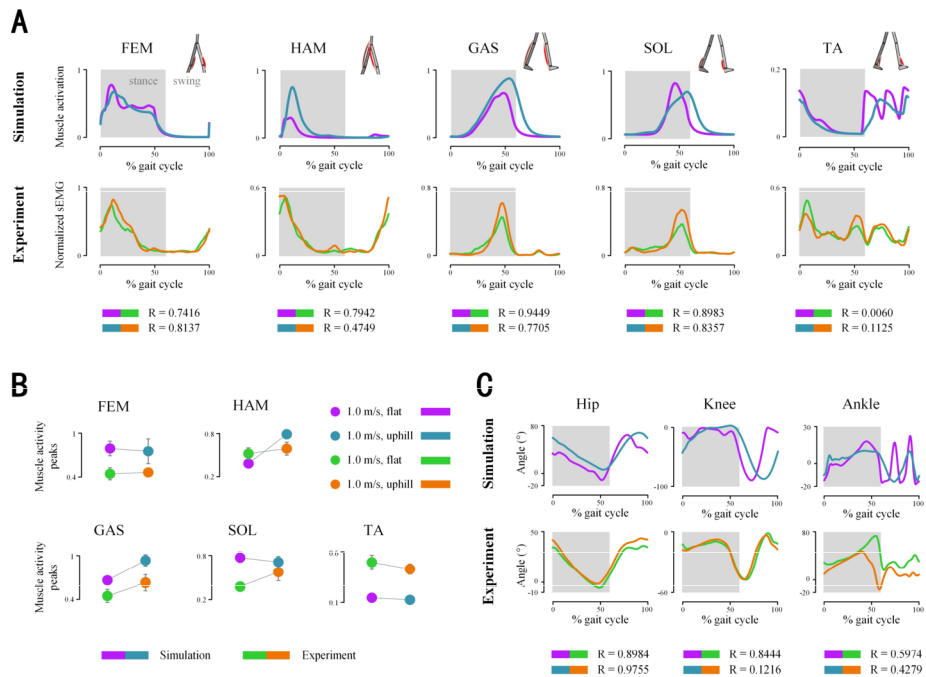


Fig. 7 Comparison between simulation and experiment results when walking flat and uphill at a speed of 1.0 m/s. (A) Activation of FEM, HAM, GAS, SOL, and TA muscles, including both model-predicted muscle activations and experimental sEMG signals. The curves are averaged results of 7 walking cycles, and the coefficients of correlation (R) between simulated and experimental data are provided. (B) The muscle activation and sEMG peaks (average $\pm$ std) of the respective muscles when walking on the two terrains. (C) Angles of the hip, knee, and ankle joints, including both model predictions and experimental data captured by inertial motion sensors. The curves are averaged results of 7 walking cycles, and the coefficients of correlation (R) between simulated and experimental data are provided (Color figure online)

4 Case study 2: walking at various speeds

In the second case, we further examine the capability of our dynamic simulation framework to capture changes in walking speed from 0.9 m/s to 1.1 m/s. The simulation and experiment testbeds are identical to those of case 1. The exoskeleton also operates in passive mode in this case.

4.1 Simulation analysis

The effectiveness of our dynamic simulation framework is further demonstrated by its ability to capture and represent the performance of the human-exoskeleton system under different walking speeds. In this case study, we performed simulations of the human-exoskeleton model walking on the flat ground at three walking speed constraints ($v_{COM}^{des} = 0.9, 1.0$, and 1.1 m/s) with all other simulation conditions in correspondence with case 1. For $v_{COM}^{des} = 0.9$ and 1.1 m/s scenarios, the muscle reflex parameters are optimized via the 2-step CMA-ES algorithm, using the outcome from the parameters of $v_{COM}^{des} = 1.0$ m/s flat walking scenario that are optimized early in the first step of case 1. Both scenarios finish the step-1 optimization in the first iteration, and it takes 16 and 20 more iterations in the second step for

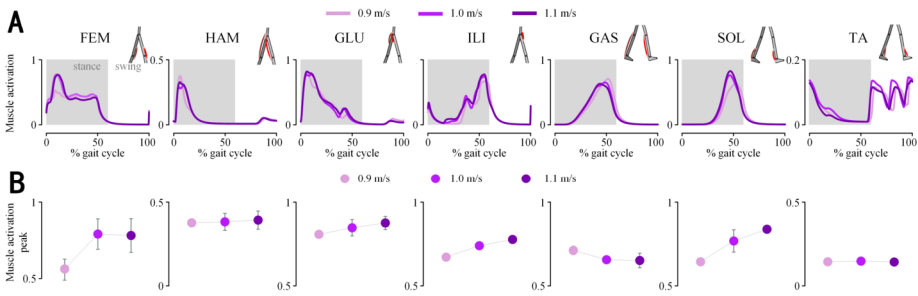


Fig. 8 Muscle activations in the right leg with respect to different walking speeds. (A) From the left to the right, showing the muscle activations of the FEM, HAM, GLU, ILI, GAS, SOL, and TA muscles when walking at speeds of 0.9, 1.0, and 1.1 m/s on the flat ground. The curves are averaged results of 7 walking cycles. (B) The muscle activation peaks (average $\pm$ std) of the respective muscles at the three walking speeds (Color figure online)

$v_{\text{COM}}^{\text{des}} = 0.9$ and 1.1 m/s, respectively. The optimized muscle reflex parameters for each scenario are provided in S8 of the Supplementary Material.

Walking speeds are considered as constraints in the optimization problem for this case. The simulation results are vividly demonstrated in Supplementary Video S2. Simulation results also reveal that while the walking speed grows from 0.9 m/s to 1.1 m/s, in addition to increases in step frequency (from 100.44 to 101.81) and in step length (from 0.5 m to 0.59 m), significant changes in muscle activations occur. Fig. 8 provides a representation of the effects of the walking speed on muscle activations. As the walking speed increases from 0.9 to 1.1 m/s, the SOL and ILI muscles, which play crucial roles in propulsion during forward walking, display a positive correlation between peak activation values and walking speeds ($p < 0.05$). The positive correlation between the activation of the FEM muscle and walking speeds can be attributed to its primary role in maintaining knee joint stiffness during the stance phase. This function is vital for supporting the body in an upright position and providing stability during the weight-bearing phase of the gait cycle. As the walking speed increases, there is a greater demand for force generation from the FEM muscle to effectively fulfill this stabilizing role, resulting in the observed positive correlation between its activation and walking speeds. Conversely, it is observed that the peak activation of the GAS muscle decreases with higher walking speeds. This may be attributed to the compensatory behavior from the SOL muscle, which has a similar function to the GAS muscle in the plantar flexor; as the walking speed increases, the activation level of the SOL muscle significantly rises, resulting in a decrease in the activation level of GAS. Notably, the peak activation levels of the GLU and HAM muscles, contributing to gait stability, do not exhibit significant changes with increasing speed. Additionally, the peak activation of the TA muscle remains relatively low and stable as walking speed increases. This phenomenon can be attributed to the influence of step length on TA muscle activation. This may be due to the tendency in simulations that the foot elevation and clearance is achieved more through the hip and knee joints, thus making the involvement of the FEM and ILI muscles more significant, while the involvement of TA muscle is lower and does not vary significantly relative to walking speeds.

4.2 Experimental validation

We conducted experiments with one subject wearing the passive hip exoskeleton, who walked on a treadmill at the speed of 0.9, 1.0, and 1.1 m/s for 1 minute each. The sub-

ject, experimental setup, and data processing remains the same as those in case 1. Fig. 9(A) demonstrates the muscle activations generated by the proposed simulation framework at speeds of 0.9, 1.0, and 1.1 m/s, which are compared to the sEMG signals measured during our experiments. It should be noted that sEMG signals and muscle activation are two distinct quantities, and their absolute values cannot be directly compared. Instead, our focus is on observing their patterns of change. In this study, with the exception of the TA muscles (to be discussed in Sect. 4), there is a strong consistency ($R > 0.6375$) between muscle activations in simulations and sEMG signals in experiments. Fig. 9(B) depicts the relationship between peak muscle activity and walking speed in both simulations and experiments. The simulation and experimental data exhibit good agreement in terms of the qualitative variation pattern of peak muscle activity with respect to walking speeds. Specifically, the results indicate that the FEM and SOL muscles demonstrate a notable increase in peak muscle activity as walking speed increases, favorable to support and propel, respectively. The HAM and TA muscles do not exhibit a significant correlation between peak muscle activities and walking speeds. Simulation results indicated a decrease in the peak activation of GAS muscles with increasing walking speed, while the decline in experimental data is not significant. This decrease in peak GAS muscle activity can be attributed to the compensatory behavior between the GAS and SOL muscles.

In addition, Fig. 9(C) demonstrates the consistency between simulation and experiment regarding joint angles ($R > 0.5974$). Note that there are some quantitative discrepancies, particularly in the knee joint angles, which can be attributed to the differences in gait characteristics among individuals. Importantly, our simulation results align with the experimental observations of Wang et al. [50], specifically concerning the knee joint angles.

Overall, the comparison between experimental and simulation results fully demonstrates the ability of our dynamic model and simulation framework to accurately predict muscle activity and kinematics. Importantly, our model is able to generate rich and informative outputs without relying on experimental tests or sensor data. This feature instills confidence in utilizing this dynamics simulation framework for studying the neuromusculoskeletal dynamics of the coupled human-exoskeleton system across various environments and gait patterns.

5 Case study 3: walking with/without exoskeleton assistance

The third case investigates the impact of exoskeleton assistance on muscle activities while the system is walking on the flat ground. The case involves two simulation scenarios: one with the hip exoskeleton in passive mode, and one with the cable-driven hip exoskeleton that provides 20 N·m of peak torque assistance. Both scenarios use the human-exoskeleton model walking on the flat ground at the speed of 1.0 m/s. All other simulation setups are identical to those of case 1 and 2.

5.1 Assistive torque profile

The assistive torque is described as a single peak curve to mimic the hip extension torque of human walking [78]. Figure 10(A) demonstrates the device operation during the gait cycle. Starting from the landing phase (90% of the gait cycle), the assistive torque is applied to the thigh brace of the hip exoskeleton through a cable. The cable is threaded with a pulley and conveys backward force that is generated by the motor positioned on the human's back. The assistive torque attains its maximum value in the early stance phase (5% of the next gait cycle), coinciding with the peak extension torque of the hip joint, to assist in bearing the

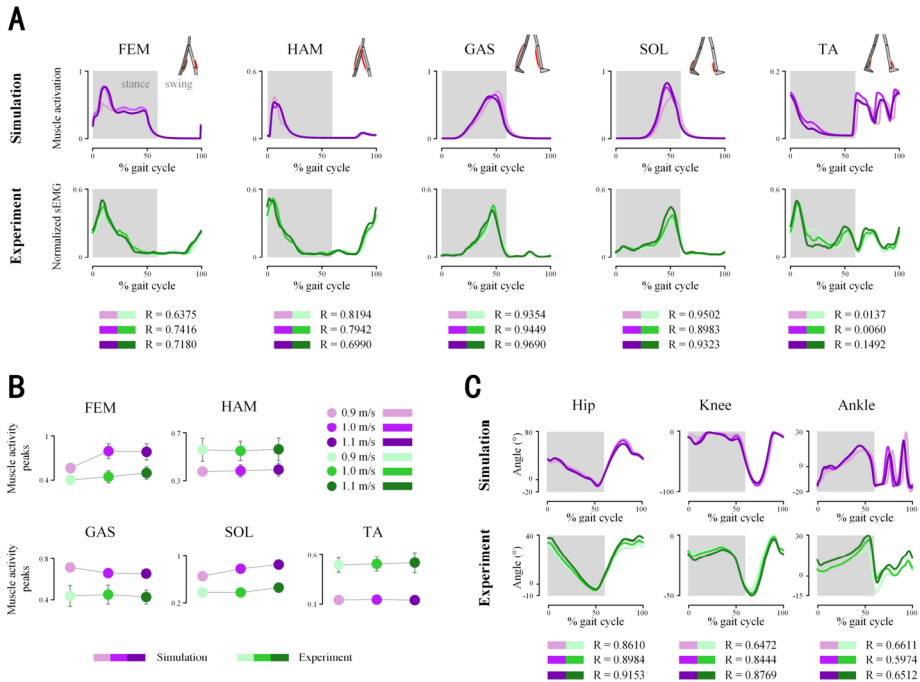


Fig. 9 Comparison between simulation and experiment results when walking at speeds of 0.9, 1.0, and 1.1 m/s on the flat ground. (A) Activation of FEM, HAM, GAS, SOL, and TA muscles, including both model-predicted muscle activations and experimental sEMG signals. The curves are averaged results of 7 walking cycles, and the coefficients of correlation (R) between the simulated and experimental data are provided. (B) The muscle activation and sEMG peaks (average $\pm$ std) of the respective muscles at the three walking speeds. (C) Angles of the hip, knee, and ankle joints, including both model predictions and experimental data captured by inertial motion sensors. The curves are averaged results of 7 walking cycles, and the coefficients of correlation (R) between simulated and experimental data are provided (Color figure online)

leg load after heel strike. After that, the assistive torque gradually decreases to 0 in the late stance phase (30% of the next gait cycle).

Figure 10(B) shows the assistive torque normalized in one gait cycle, defined as a single peak curve realized by using two cubic splines joint at their peaks. We use four parameters to describe the curve: start time, peak time, duration time, and peak torque. The first three temporal-related parameters are normalized to gait cycle and are denoted as X%. In the scenario where the exoskeleton provides assistance, the four parameters are assigned as 90%, 15%, 40% and 20 N·m, respectively, referencing the human walking biomechanics [56].

5.2 Simulation analysis

In this case study, we compare the simulation results of the human-exoskeleton model walking with/without exoskeleton assistance on the flat ground at $v_{COM}^{des} = 1.0$ m/s. For the exoskeleton-assisting scenario, the muscle reflex parameters of the system are optimized through CMA-ES algorithm, using the outcome from the parameters of the $v_{COM}^{des} = 1.0$ m/s flat walking scenario optimized in the first step of case 1. It took 8 iterations to finish the first step of the optimization, and 35 more iterations to finish the second step. The opti-

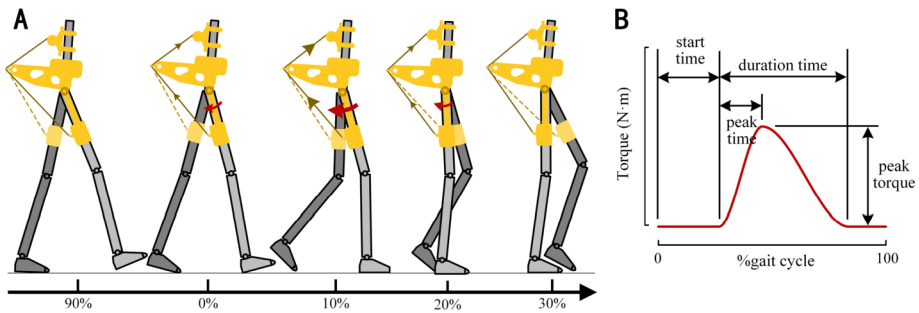


Fig. 10 Assistive torque profile of the exoskeleton. (A) Overview of the device operation during the gait cycle. The assistive extension torque starts from the landing phase (90% of the gait cycle), peaks in the early stance phase (5% of the next gait cycle), and ends in the late stance phase (30% of the gait cycle). (B) The parameterized assistive torque curve, determined by start time, peak time, duration time and peak torque (Color figure online)

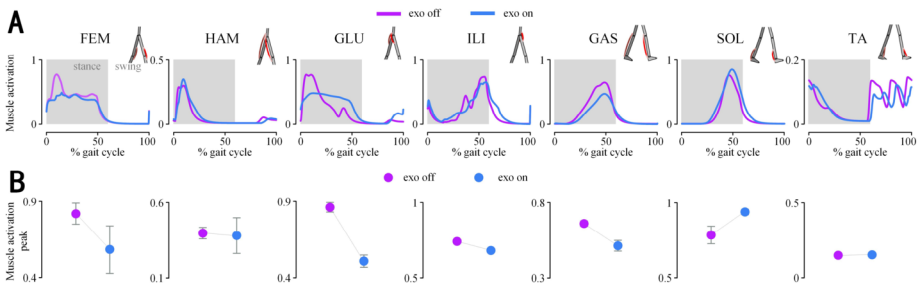


Fig. 11 Muscle activations in the right leg with the exoskeleton providing no assistance (exo off) or providing 20 N·m of peak assistance torque (exo on). (A). From the left to the right, showing the muscle activations of the FEM, HAM, GLU, ILI, GAS, SOL, and TA muscles when walking at speeds of 1.0 m/s on the flat ground. The curves are averaged results of 7 walking cycles. (B) The muscle activation peaks (average $\pm$ std) of the respective muscles in walking with exoskeleton off and on (Color figure online)

mized muscle reflex parameters for each scenario are provided in S8 of the Supplementary Material, and the simulation results are vividly demonstrated in Supplementary Video S3

According to the simulation results, the scenario with exoskeleton assistance yields an overall muscle metabolic expenditure of 4.14, whereas the scenario without assistance yields 4.52. This implies a 8.4% decrease in muscle effort due to the exoskeleton assistance. Moreover, Fig. 11 demonstrates the effect of the hip exoskeleton assistance on the muscle activation patterns. Notably, the activation of the FEM and GLU muscles decreases the most significantly after the exoskeleton provides assistance ($p < 0.05$). The functional substitution of the GLU muscle by the hip exoskeleton accounts for the decrease in its activation. The hip exoskeleton primarily delivers extension torque to the hip joint in the landing and the early stance phase of the gait cycle, which coincides the role of the GLU muscle, and also results in less requirement for the FEM muscle to maintain the knee stiffness after heel strike. The HAM and TA muscles remain relatively unchanged in the two scenarios ($p > 0.05$), which could be explained by the negligible influence of the hip extension assistance torque on the knee flexion and ankle plantarflexion. Moreover, we notice a significant decrease in the activation of the ILI and GAS muscles, which is unexpected. This may be due to the fact that the emergence of exoskeleton assistance altered the muscle contraction pref-

erence of the human, and consequently facilitated the simulation framework to discover a more energy-saving walking pattern. The SOL muscle also exhibits compensatory behavior to the GAS muscle, as they both share similar functions in ankle plantarflexion.

6 Discussion

This study presents an optimization-based comprehensive dynamic simulation framework for the thorough analysis of human-exoskeleton coupling systems. The framework integrates the human neuromusculoskeletal system with exoskeletons and incorporates adjustable human-exoskeleton coupling and foot-ground contact. It enables the prediction and analysis of joint kinematics of the coupled exoskeleton-human system in different walking conditions and environments, while also integrating a neuromusculoskeletal feedback loop for biomechanical analysis. The introduced simulation framework possesses the capability to comprehensively simulate the multibody dynamics of the coupled human-exoskeleton system in different scenarios without relying on experimental data. The proposed simulation framework also offers valuable insights into the interaction forces between humans and exoskeletons, which are crucial for assessing the effectiveness and safety of exoskeletons in enhancing human movement. However, it is important to acknowledge certain limitations and minor discrepancies in the quantitative aspect of some results. The succeeding subsections will offer a detailed discussion of these topics by delving into their specifics.

6.1 Advantages and limitations

Notably, the proposed simulation framework employs an optimization-based approach to determine multi-source biomechanical signals, including muscle activation, muscle forces, joint torques, and joint kinematics, without relying on experimental data. Rather than utilizing empirical preset values or direct optimization, these signals are obtained by optimizing the muscle reflex parameters, considering various interactions such as muscle activities, muscle contraction dynamics, human-exoskeleton interaction, and system-environment interaction. This approach, along with improved segmentation of the gait phase, improves the accuracy of muscle excitation, enhances the representation of physiological behavior, and offers a realistic depiction of human movement within the dynamic simulation context.

The simulation of the coupled neuromusculoskeletal-exoskeletal system in this framework is achieved through a well-designed optimization strategy. The optimization process consists of two steps, each serving a specific purpose. In the first step, our primary goal is to maximize the walking distance without falls. The focus is on ensuring that the model exhibits reasonable and feasible walking behavior. In the second step, a multidimensional evaluation criterion is employed, incorporating factors such as walking speed, joint angle constraints, and muscle energy consumption. This criterion is employed to guide the optimization process and constrain the model's walking behavior, aiming for a more natural and human-like gait pattern.

The developed simulation framework in this study capturing the dynamic response of the coupled human-exoskeleton system across a range of walking speeds and environmental conditions. This can be attributed to the incorporation of a neuro-musculoskeletal closed loop, which is achieved through the optimization of muscle reflex parameters using multidimensional evaluation criteria. The optimization process ensures that important constraints related to muscle-tendon force balance, rigid body dynamics, joint angle limits, and the

boundaries of muscle excitation and activation are consistently met. Moreover, the variations in environment and walking speed are appropriately accounted for in the optimization objectives and the adjustable foot-ground contact forces within the rigid body dynamics. These factors play a key role in shaping the neuromusculoskeletal closed-loop and subsequently influence the kinematic and biomechanical outcomes of the coupled system.

While our method offers several notable advantages, it is important to acknowledge the limitations of this study. One limitation is the relatively low computational efficiency of the simulation framework, albeit a two-step optimization strategy and parallel optimization with the CMA-ES algorithm have been used. This is an important consideration, as it may limit the applicability of the framework in certain real-time or time-sensitive applications. To address this limitation and improve computational efficiency, further research and development efforts could explore alternative optimization algorithms or parallel computing techniques, which could help reduce the computational burden and enable faster simulations without compromising the accuracy and reliability of the results. Additionally, better efficiency would be achieved if the simulation framework is deployed in platform developed in languages like C/C++. Efforts to enhance computational efficiency would not only make the simulation framework more practical for time-sensitive applications but also expand its potential applications in areas such as real-time control and virtual reality simulations.

Another limitation of this study is the presence of individual differences in gait characteristics and the difficulty in accurately measuring certain parameters of the human muscle model. These factors can induce quantitative differences between the simulation results and experimental data. While exact numerical agreement between simulation and experimental results cannot be met or pursued, the overall behavior and trends exhibited by the system remain consistent. This qualitative consistency allows for valuable insights into the dynamics and interactions of the human-exoskeleton system, aiding in understanding its performance and improving its design. In addition, we acknowledge that the sample size in this study is confined to a single subject who repeatedly performed the gait tests (as the exoskeleton needs to be customized to the subject). While this approach has its limitations due to the inherent variability among individuals, it nevertheless provides valuable insights. With the exception of certain muscles, our experiment results are consistent with broader patterns observed in similar studies [79, 80], suggesting that the outcomes of our dynamic simulation framework hold relevance to general gait behaviors. However, future studies may benefit from a larger sample size and a wider range of experimental validation, including different populations performing various activities in different environments. This would help to refine the modeling and simulation approaches and enhance the applicability of the framework. Besides, in case 3, the assessment of reflex-based model predictions for exoskeleton assistance is conducted through simulations, without accompanying experimental trials. Recognizing this, future efforts will be directed towards incorporating experimental studies, aiming to complement and substantiate the simulation findings.

Lastly, it should be noted that our optimization method does not provide a guarantee of convergence to the global optimum. Instead, it may converge to different local optima depending on the specific problem and initial conditions. However, after a sufficient number of iterations, our optimization method can eventually converge to a similar local optimal solution that effectively characterizes human motion within the constraints and objectives defined in the optimization problem. While it may not be the absolute global optimum, the obtained solution captures the essential characteristics and behaviors of human movement. To mitigate the impact of local optima, it is crucial to carefully design the optimization problem by incorporating appropriate constraints, objectives, and evaluation criteria.

6.2 Result interpretation and error analysis

Using the dynamic simulation framework proposed in this study, we have conducted a set of simulations and experiments to analyze the gait of the coupled human-exoskeleton system under different terrains and walking speeds. The results obtained from these investigations demonstrate that the proposed framework successfully captures the patterns of joint kinematics and muscle activity during walking across a wide range of speeds and environmental conditions. Notably, the simulated patterns align with the trends observed in experimental data, providing evidence for the effectiveness of the dynamic simulation framework. This validation highlights the framework's capability to faithfully reproduce and predict the behavior of the human-exoskeleton system during walking, establishing it as a valuable tool for further research in this field.

The simulation framework presented in this study can accurately capture the functional variations in muscle activity during different walking conditions. Specifically, it effectively represents the dynamic changes in the activity of different muscles based on their specific functions. For instance, the GLU and HAM muscles, which contribute to walking stability, exhibit no significant correlation between peak muscle activities and walking speeds. However, during uphill walking, there is a notable increase in the peak activity of these muscles, indicating their role in providing stability in changing terrains. The FEM muscle, responsible for support, shows a positive correlation with walking speed, reflecting its involvement in generating the necessary support for faster walking. Interestingly, during uphill walking, where a shorter step length promotes stability, no significant correlation is observed. Moreover, the proposed framework successfully captures the phenomenon of muscle compensation between muscles with similar functions, as evidenced by the interactions between the SOL and GAS muscles. This ability to represent such muscle interactions and compensatory mechanisms is achieved by incorporating a closed neuromusculoskeletal feedback loop, coupled with a rigid-body dynamics model of the human-exoskeleton system. The solution of this coupled system is obtained by optimization, considering factors such as velocity, stability, energy, and other objectives. While the obtained solution may not be globally optimal, it physiologically demonstrates the variation pattern of different muscle activities relative to walking conditions.

While the proposed framework demonstrates its effectiveness in qualitatively capturing the kinematics and biomechanics during walking with exoskeletons, it is important to acknowledge the existence of quantitative disparities between the simulation results and the experimental data, particularly with regard to the activity of the TA muscle. These disparities can be attributed to several factors, including individual differences in gait characteristics and limitations in the modeling of the muscle's driving mechanism in the simulation. In our model, the excitation of the TA muscle is determined by the positive feedback of the muscle length and the negative feedback of the activity of the SOL muscle. However, the actual function of the TA muscle during walking is more complex and multifaceted. In reality, the TA muscle serves as the primary dorsiflexor of the ankle joint, requiring a larger force generation after heel strike to maintain the ankle joint in a neutral position and prevent foot slapping. Additionally, it needs to contract during the swing phase to ensure proper foot clearance. These dynamic functions of the TA muscle are not fully captured in our current model, leading to noticeable differences between the simulated and actual TA muscle activities. Furthermore, the synergistic and antagonistic interactions between TA muscle and other muscles involved in ankle joint control may be more intricate than our current understanding. The complexity of these interactions and the precise coordination among muscles during the walking process present challenges in accurately modeling and simulating their

behavior. The simplistic muscle reflex mechanism of the TA muscle also leads to disparities in ankle movement between the simulation results and the experimental data, which achieves a relatively low R-value at $0.4 \sim 0.7$. Furthermore, we observed discrepancies between the simulated and experimental results for the angles of the hip and knee joints. Such mismatches could partly arise from the not yet fully comprehensive representation of muscles, for example, the vastus lateralis, vastus intermedius and vastus medialis that play a role in active knee extension during the swing phase of walking are not considered [42]. This limitation could lead to a tendency towards a higher-stepping gait pattern, as indicated by an elevated hip. We acknowledge that a more comprehensive refinement and enhancement of muscle dynamics within the model could contribute to a more accurate replication of the movements inherent in human gait. Future work will focus on this area, aiming to improve the model's accuracy and reliability in simulating the biomechanical effects when exoskeletons are used.

While the proposed framework provides valuable insights into the overall dynamics of the human-exoskeleton system and captures qualitative trends in muscle activity, the discrepancies observed in the TA muscle activity highlight the need for further refinement and development of the muscle modeling approach. Future research efforts can focus on incorporating more detailed and comprehensive muscle models that better capture the intricate dynamics and interactions among muscles, leading to a more accurate representation of muscle activity during walking with exoskeletons.

7 Conclusions

This study advances the field of human-exoskeleton interaction by presenting a focused examination of the integration of exoskeleton models into a refined reflex-based gait model. At its core, our work seeks to understand the dynamic responses of subjects to varying levels of assistance provided by exoskeletons, facilitated through a comprehensive simulation framework. The simulation framework encompasses essential components including human lower limb muscle reflex mechanisms, muscle activation dynamics, muscle-tendon contraction dynamics, musculoskeletal dynamics, human body-exoskeleton interaction, and foot-ground contact. Notably, the simulation framework does not rely on external measurements or empirical data but instead employs an optimization-based approach that considers multi-objective fitness functions. This allows for a thorough analysis of the kinematic and biomechanical characteristics of the coupled human-exoskeleton system under diverse walking and environmental conditions. By comparing the simulation results with experimental data, the study demonstrates the convenience and effectiveness of the proposed framework in capturing the qualitative neuromuscular reflex patterns and joint kinematics observed in individuals wearing exoskeletons during walking across different scenarios. The walking simulation of the human-exoskeleton coupled system with or without exoskeleton assistance further deepens the understanding of the influence of exoskeleton assistance on human muscle activity. While certain limitations exist, they pave the way for future research directions in the dynamic simulation of human-exoskeleton coupling.

Furthermore, the model-based simulation framework presented in this study provides a wealth of information that facilitates efficient and cost-effective performance testing of the human-exoskeleton system under various environmental and gait conditions. This capability is particularly advantageous for integrating and evaluating novel or updated exoskeleton designs and control strategies. By eliminating the need for frequent prototyping of new exoskeletons and conducting expensive human experiments, the simulation framework would

streamline the development and evaluation process, leading to significant time and cost savings.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Ethical approval The protocol received approval from the Ethical Committee of Fudan University (protocol code FE21124, approved on 16 August 2021).

Consent to participate Written informed consent was obtained from the subject involved in the study.

Competing interests The authors declare no competing interests.

References

1. Dollar, A.M., Herr, H.: Lower extremity exoskeletons and active orthoses: challenges and state-of-the-art. *IEEE Trans. Robot.* **24**, 144–158 (2008). <https://doi.org/10.1109/TRO.2008.915453>
2. Harant, M., Näf, M.B., Mombaur, K.: Multibody dynamics and optimal control for optimizing spinal exoskeleton design and support. *Multibody Syst. Dyn.* **57**, 389–411 (2023). <https://doi.org/10.1007/s11044-023-09877-w>
3. Nasr, A., Bell, S., McPhee, J.: Optimal design of active-passive shoulder exoskeletons: a computational modeling of human-robot interaction. *Multibody Syst. Dyn.* **57**, 73–106 (2023). <https://doi.org/10.1007/s11044-022-09855-8>
4. Nam, K.Y., Kim, H.J., Kwon, B.S., Park, J.W., Lee, H.J., Yoo, A.: Robot-assisted gait training (Lokomat) improves walking function and activity in people with spinal cord injury: a systematic review. *J. NeuroEng. Rehabil.* **14**, 1–13 (2017). <https://doi.org/10.1186/s12984-017-0232-3>
5. de Looze, M.P., Bosch, T., Krause, F., Stadler, K.S., O’Sullivan, L.W.: Exoskeletons for industrial application and their potential effects on physical work load. *Ergonomics* **59**, 671–681 (2016). <https://doi.org/10.1080/00140139.2015.1081988>
6. Collins, S.H., Bruce Wiggin, M., Sawicki, G.S.: Reducing the energy cost of human walking using an unpowered exoskeleton. *Nature* **522**, 212–215 (2015). <https://doi.org/10.1038/nature14288>
7. Ishmael, M.K., Archangeli, D., Lenzi, T.: Powered hip exoskeleton improves walking economy in individuals with above-knee amputation. *Nat. Med.* **27**, 1783–1788 (2021). <https://doi.org/10.1038/s41591-021-01515-2>
8. Park, K.W., Choi, J., Kong, K.: Iterative learning of human behavior for adaptive gait pattern adjustment of a powered exoskeleton. *IEEE Trans. Robot.* **38**, 1395–1409 (2022). <https://doi.org/10.1109/TRO.2022.3144955>

9. Ishmael, M.K., Archangeli, D., Lenzi, T.: A powered hip exoskeleton with high torque density for walking, running, and stair ascent. *IEEE/ASME Trans. Mechatron.* **27**, 4561–4572 (2022). <https://doi.org/10.1109/TMECH.2022.3159506>
10. Sarkisian, S.V., Ishmael, M.K., Lenzi, T.: Self-aligning mechanism improves comfort and performance with a powered knee exoskeleton. *IEEE Trans. Neural Syst. Rehabil. Eng.* **29**, 629–640 (2021). <https://doi.org/10.1109/TNSRE.2021.3064463>
11. Zoss, A.B., Kazerooni, H., Chu, A.: Biomechanical design of the Berkeley Lower Extremity Exoskeleton (BLEEX). *IEEE/ASME Trans. Mechatron.* **11**, 128–138 (2006). <https://doi.org/10.1109/TMECH.2006.871087>
12. Martini, E., Crea, S., Parri, A., Bastiani, L., Faraguna, U., McKinney, Z., Molino-Lova, R., Pratali, L., Vitiello, N.: Gait training using a robotic hip exoskeleton improves metabolic gait efficiency in the elderly. *Sci. Rep.* **9**, 7157 (2019). <https://doi.org/10.1038/s41598-019-43628-2>
13. Mo, F., Zhang, Q., Zhang, H., Long, J., Wang, Y., Chen, G., Ye, J.: A simulation-based framework with a proprioceptive musculoskeletal model for evaluating the rehabilitation exoskeleton system. *Comput. Methods Programs Biomed.* **208**, 106270 (2021). <https://doi.org/10.1016/j.cmpb.2021.106270>
14. Padmanabha, G.A., Ramalingasetty, S.T., Vetrivel, B., Mukherjee, I., Omkar, S.N., Sivakumar, R.: Computational modelling of musculoskeletal to predict the human response with exoskeleton suit. *Int. J. Biomechanics Biomed. Robot.* **3**, 169 (2020). <https://doi.org/10.1504/ijbbr.2020.108441>
15. Gordon, D.F.N., McGreavy, C., Christou, A., Vijayakumar, S.: Human-in-the-loop optimization of exoskeleton assistance via online simulation of metabolic cost. *IEEE Trans. Robot.* **38**, 1410–1429 (2022). <https://doi.org/10.1109/TRO.2021.3133137>
16. Ezati, M., Ghannadi, B., McPhee, J.: A review of simulation methods for human movement dynamics with emphasis on gait. *Multibody Syst. Dyn.* **47**, 265–292 (2019). <https://doi.org/10.1007/s11044-019-09685-1>
17. Bianco, N.A., Franks, P.W., Hicks, J.L., Delp, S.L.: Coupled exoskeleton assistance simplifies control and maintains metabolic benefits: a simulation study. *PLoS ONE* **17**, 1–18 (2022). <https://doi.org/10.1371/journal.pone.0261318>
18. Li, X., Chen, J., Wang, W., Zhang, F., Han, H., Zhang, J.: Using predictive simulation methods to design suitable assistance modes for human walking on slopes. 2020 3rd Int. Conf. Control Robot. ICCR **2020**, 169–175 (2020). <https://doi.org/10.1109/ICCR51572.2020.9344320>
19. Uhlrich, S.D., Uchida, T.K., Lee, M.R., Delp, S.L.: Ten steps to becoming a musculoskeletal simulation expert: a half-century of progress and outlook for the future. *J. Biomech.* **154**, 1–41 (2023). <https://doi.org/10.1016/j.jbiomech.2023.111623>
20. Yu, J., Zhang, S., Wang, A., Li, W.: Human gait analysis based on OpenSim. In: International Conference on Advanced Mechatronic Systems, ICAMechS, pp. 278–281 (2020)
21. Geyer, H., Herr, H.: A muscle-reflex model that encodes principles of legged mechanics produces human walking dynamics and muscle activities. *IEEE Trans. Neural Syst. Rehabil. Eng.* **18**, 263–273 (2010). <https://doi.org/10.1109/TNSRE.2010.2047592>
22. Sinica, A.M.: A forward-inverse dynamics modeling framework for a forward-inverse dynamics modeling framework for human musculoskeletal multibody system (2022)
23. Ignasiak, D.: A novel method for prediction of postoperative global sagittal alignment based on full-body musculoskeletal modeling and posture optimization. *J. Biomech.* **102**, 109324 (2020). <https://doi.org/10.1016/j.jbiomech.2019.109324>
24. Karatsidis, A., Jung, M., Schepers, H.M., Bellusci, G., de Zee, M., Veltink, P.H., Andersen, M.S.: Musculoskeletal model-based inverse dynamic analysis under ambulatory conditions using inertial motion capture. *Med. Eng. Phys.* **65**, 68–77 (2019). <https://doi.org/10.1016/j.medengphy.2018.12.021>
25. Mathai, B., Gupta, S.: Numerical predictions of hip joint and muscle forces during daily activities: a comparison of musculoskeletal models. *Proc. Inst. Mech. Eng., H J. Eng. Med.* **233**, 636–647 (2019). <https://doi.org/10.1177/0954411919840524>
26. Millard, M., Emonds, A.L., Harant, M., Mombaur, K.: A reduced muscle model and planar musculoskeletal model fit for the simulation of whole-body movements. *J. Biomech.* **89**, 11–20 (2019). <https://doi.org/10.1016/j.jbiomech.2019.04.004>
27. Weng, J., Hashemi, E., Arami, A.: Adaptive reference inverse optimal control for natural walking with musculoskeletal models. *IEEE Trans. Neural Syst. Rehabil. Eng.* **30**, 1567–1575 (2022). <https://doi.org/10.1109/TNSRE.2022.3180690>
28. Willson, A.M., Anderson, A.J., Richburg, C.A., Muir, B.C., Czerniecki, J., Steele, K.M., Aubin, P.M.: Full body musculoskeletal model for simulations of gait in persons with transtibial amputation. *Comput. Methods Biomed. Eng.* **26**, 412–423 (2023). <https://doi.org/10.1080/10255842.2022.2065630>
29. Quental, C., Simões, F., Sequeira, M., Ambrósio, J., Vilas-Boas, J.P., Nakashima, M.: A multibody methodological approach to the biomechanics of swimmers including hydrodynamic forces. *Multibody Syst. Dyn.* **57**, 413–426 (2023). <https://doi.org/10.1007/s11044-022-09865-6>

30. Engelhardt, L., Melzner, M., Havelkova, L., Fiala, P., Christen, P., Dendorfer, S., Simon, U.: A new musculoskeletal AnyBody™ detailed hand model. *Comput. Methods Biomech. Biomed. Eng.* **24**, 777–787 (2020). <https://doi.org/10.1080/10255842.2020.1851367>
31. Fonk, R., Schneeweiss, S., Simon, U., Engelhardt, L.: Hand motion capture from a 3d leap motion controller for a musculoskeletal dynamic simulation. *Sensors* **21**, 1199 (2021). <https://doi.org/10.3390/s21041199>
32. Michnik, R., Zadoń, H., Nowakowska-Lipiec, K., Jochymczyk-Woźniak, K., Myśliwiec, A., Mitas, A.W.: The effect of the pelvis position in the sagittal plane on loads in the human musculoskeletal system. *Acta Bioeng. Biomech.* **22**, 33–42 (2020). <https://doi.org/10.37190/ABB-01606-2020-02>
33. Trinler, U., Schwameder, H., Baker, R., Alexander, N.: Muscle force estimation in clinical gait analysis using AnyBody and OpenSim. *J. Biomech.* **86**, 55–63 (2019). <https://doi.org/10.1016/j.jbiomech.2019.01.045>
34. Ackermann, M., van den Bogert, A.J.: Optimality principles for model-based prediction of human gait. *J. Biomech.* **43**, 1055–1060 (2010). <https://doi.org/10.1016/j.jbiomech.2009.12.012>
35. Caruntu, D.I., Moreno, R.: Human knee inverse dynamics model of vertical jump exercise. *J. Comput. Nonlinear Dyn.* **14**, 101005 (2019). <https://doi.org/10.1115/1.4044246>
36. Zhao, Y., Li, Z., Zhang, Z., Qian, K., Xie, S.: An EMG-driven musculoskeletal model for estimation of wrist kinematics using mirrored bilateral movement. *Biomed. Signal Process. Control* **81**, 104480 (2023). <https://doi.org/10.1016/j.bspc.2022.104480>
37. Yao, S., Zhuang, Y., Li, Z., Song, R.: Adaptive admittance control for an ankle exoskeleton using an EMG-driven musculoskeletal model. *Front. Neurorobot.* **12** (2018). <https://doi.org/10.3389/fnbot.2018.00016>
38. Sun, L., An, H., Ma, H., Wei, Q.: Research on assistance evaluation and online optimization method of lower limb exoskeleton based on musculoskeletal model. In: *Proceedings of the 31st Chinese Control and Decision Conference, CCDC 2019*, pp. 4974–4979. IEEE Press, New York (2019)
39. Moya-Esteban, A., van der Kooij, H., Sartori, M.: Robust estimation of lumbar joint forces in symmetric and asymmetric lifting tasks via large-scale electromyography-driven musculoskeletal models. *J. Biomech.* **144**, 111307 (2022). <https://doi.org/10.1016/j.jbiomech.2022.111307>
40. Bélaïse, C., Dal Maso, F., Michaud, B., Mombaur, K., Begon, M.: An EMG-marker tracking optimisation method for estimating muscle forces. *Multibody Syst. Dyn.* **42**, 119–143 (2018). <https://doi.org/10.1007/s11044-017-9587-2>
41. Guo, J., Chen, J., Wang, J., Ren, G., Tian, Q., Guo, C.: EMG-assisted forward dynamics simulation of subject-specific mandible musculoskeletal system. *J. Biomech.* **139**, 111143 (2022). <https://doi.org/10.1016/j.jbiomech.2022.111143>
42. Song, S., Geyer, H.: A neural circuitry that emphasizes spinal feedback generates diverse behaviours of human locomotion. *J. Physiol.* **593**, 3493–3511 (2015). <https://doi.org/10.1113/JP270228>
43. Van Wouwe, T., Ting, L.H., De Groote, F.: An approximate stochastic optimal control framework to simulate nonlinear neuromusculoskeletal models in the presence of noise. *PLoS Comput. Biol.* **18**, 1–30 (2022). <https://doi.org/10.1371/journal.pcbi.1009338>
44. Dorschky, E., Nitschke, M., Seifer, A.K., van den Bogert, A.J., Eskofier, B.M.: Estimation of gait kinematics and kinetics from inertial sensor data using optimal control of musculoskeletal models. *J. Biomech.* **95**, 109278 (2019). <https://doi.org/10.1016/j.jbiomech.2019.07.022>
45. Falisse, A., Serrancó, G., Dembia, C.L., Gillis, J., Jonkers, I., De Groote, F.: Rapid predictive simulations with complex musculoskeletal models suggest that diverse healthy and pathological human gaits can emerge from similar control strategies. *J. R. Soc. Interface* **16**, 20190402 (2019). <https://doi.org/10.1098/rsif.2019.0402>
46. Shourijeh, M.S., McPhee, J.: Forward dynamic optimization of human gait simulations: a global parameterization approach. *J. Comput. Nonlinear Dyn.* **9**, 031018 (2014). <https://doi.org/10.1115/1.4026266>
47. Hansen, N.: Reducing the time complexity of the derandomized evolution strategy with covariance matrix adaptation (CMA-ES). *Evol. Comput.* **11**, 1–18 (2003)
48. Wang, X., Chen, J., Qiao, H.: Motion learning and rapid generalization for musculoskeletal systems based on recurrent neural network modulated by initial states. *IEEE Trans. Cogn. Dev. Syst.* **14**, 1691–1704 (2022). <https://doi.org/10.1109/TCDS.2021.3136854>
49. Chen, J., Qiao, H.: Muscle-synergies-based neuromuscular control for motion learning and generalization of a musculoskeletal system. *IEEE Trans. Syst. Man Cybern. Syst.* **51**, 3993–4006 (2021). <https://doi.org/10.1109/TSMC.2020.2966818>
50. Wang, X., Guo, J., Tian, Q.: A forward-inverse dynamics modeling framework for human musculoskeletal multibody system. *Acta Mech. Sin. Xuebao* **38**, 522140 (2022). <https://doi.org/10.1007/s10409-022-22140-x>
51. Diao, H., Xin, H., Dong, J., He, X., Li, D., Jin, Z.: Prediction of cervical spinal joint loading and secondary motion using a musculoskeletal multibody dynamics model via force-dependent kinematics approach. *Spine* **42**(24), E1403–E1409 (2017). <https://doi.org/10.1097/BRS.0000000000002176>

52. Andersen, M.S., De Zee, M., Damsgaard, M., Nolte, D., Rasmussen, J.: Introduction to force-dependent kinematics: theory and application to mandible modeling. *J. Biomech. Eng.* **139**, 091001 (2017). <https://doi.org/10.1115/1.4037100>
53. Thelen, D.G., Anderson, F.C.: Using computed muscle control to generate forward dynamic simulations of human walking from experimental data. *J. Biomech.* **39**, 1107–1115 (2006). <https://doi.org/10.1016/j.jbiomech.2005.02.010>
54. Thelen, D.G., Anderson, F.C., Delp, S.L.: Generating dynamic simulations of movement using computed muscle control. *J. Biomech.* **36**, 321–328 (2003). [https://doi.org/10.1016/S0021-9290\(02\)00432-3](https://doi.org/10.1016/S0021-9290(02)00432-3)
55. Shourijeh, M.S., Smale, K.B., Potvin, B.M., Benoit, D.L.: A forward-muscular inverse-skeletal dynamics framework for human musculoskeletal simulations. *J. Biomech.* **49**, 1718–1723 (2016). <https://doi.org/10.1016/j.jbiomech.2016.04.007>
56. Whittle, M.W.: *Gait and Analysis an Introduction* (2007)
57. Whittle, M.W.: *Gait Analysis: An Introduction*. Butterworth-Heinemann, UK (2003)
58. Geyer, H., Seyfarth, A., Blickhan, R.: Positive force feedback in bouncing gaits? *Proc. R. Soc. Lond. B, Biol. Sci.* **270**, 2173–2183 (2003). <https://doi.org/10.1098/rspb.2003.2454>
59. Geijtenbeek, T., Van De Panne, M., Van Der Stappen, A.F.: Flexible muscle-based locomotion for bipedal creatures. *ACM Trans. Graph.* **32** (2013). <https://doi.org/10.1145/2508363.2508399>
60. Rockenfeller, R., Günther, M., Schmitt, S., Götz, T.: Comparative sensitivity analysis of muscle activation dynamics. *Comput. Math. Methods Med.* **2015** (2015). <https://doi.org/10.1155/2017/6752731>
61. Guo, J., Huang, H., Yu, Y., Liang, Z., Ambrósio, J., Zhao, Z., Ren, G., Ao, Y.: Modeling muscle wrapping and mass flow using a mass-variable multibody formulation. *Multibody Syst. Dyn.* **49**, 315–336 (2020). <https://doi.org/10.1007/s11044-020-09733-1>
62. Guo, J., Sun, Y., Hao, Y., Cui, L., Ren, G.: A mass-flowing muscle model with shape restrictive soft tissues: correlation with sonoelastography. *Biomech. Model. Mechanobiol.* **19**, 911–926 (2020). <https://doi.org/10.1007/s10237-019-01260-z>
63. Millard, M., Uchida, T., Seth, A., Delp, S.L.: Flexing computational muscle: modeling and simulation of musculotendon dynamics. *J. Biomech. Eng.* **135**, 021005 (2013). <https://doi.org/10.1115/1.4023390>
64. Charles, J.P., Cappellari, O., Spence, A.J., Hutchinson, J.R., Wells, D.J.: Musculoskeletal geometry, muscle architecture and functional specialisations of the mouse hindlimb. *PLoS ONE* **11**, e0147669 (2016). <https://doi.org/10.1371/journal.pone.0147669>
65. Geijtenbeek, T.: SCONE: open source software for predictive simulation of biological motion. *J. Open Sour. Softw.* **4**, 1421 (2019). <https://doi.org/10.21105/joss.01421>
66. Jin, W., Fang, H.: Design and preliminary evaluation of a lightweight, cable-driven hip exoskeleton for walking assistance. In: 2022 IEEE International Conference on Robotics and Biomimetics, ROBIO 2022, pp. 291–296. IEEE Press, New York (2022)
67. Mitchell, H.H., Hamilton, T.S., Steggerda, F.R., Bean, H.W.: The chemical composition of the adult human body and its bearing on the biochemistry of growth. *J. Biol. Chem.* **158**, 625–637 (1945)
68. Anderson, F.C., Pandy, M.G.: A dynamic optimization solution for vertical jumping in three dimensions. *Comput. Methods Biomech. Biomed. Eng.* **2**, 201–231 (1999). <https://doi.org/10.1080/10255849908907988>
69. Langlois, K., Rodriguez-Cianca, D., Serrien, B., De Winter, J., Verstraten, T., Rodriguez-Guerrero, C., Vanderborght, B., Lefeber, D.: Investigating the effects of strapping pressure on human-robot interface dynamics using a soft robotic cuff. *IEEE Trans. Med. Robot. Bionics.* **3**, 146–155 (2020). <https://doi.org/10.1109/TMRB.2020.3042255>
70. Ka, D.M., Hong, C., Toan, T.H., Qiu, J.: Minimizing human-exoskeleton interaction force by using global fast sliding mode control. *Int. J. Control. Autom. Syst.* **14**, 1064–1073 (2016). <https://doi.org/10.1007/s12555-014-0395-7>
71. Hunt, K., Crossley, E.: Coefficient of restitution interpreted as damping in vibroimpact. *J. Appl. Mech.* (1975)
72. Simbody (2023). HuntCrossleyForce class reference, simbody.github.io/3.7.0/classSimTK_1_1HuntCrossleyForce.html
73. Wang, J.M., Hamner, S.R., Delp, S.L., Koltun, V.: Optimizing locomotion controllers using biologically-based actuators and objectives. In: *ACM Transactions on Graphics* (2012)
74. Hansen, N.: The CMA evolution strategy: a comparing review. *Stud. Fuzziness Soft Comput.* **192**, 75–102 (2006). https://doi.org/10.1007/11007937_4
75. Buchanan, T.S., Lloyd, D.G., Manal, K., Besier, T.F.: Neuromusculoskeletal modeling: estimation of muscle forces and joint moments and movements from measurements of neural command. *J. Appl. Biomech.* **20**, 367–395 (2004)
76. Saito, A., Tomita, A., Ando, R., Watanabe, K., Akima, H.: Similarity of muscle synergies extracted from the lower limb including the deep muscles between level and uphill treadmill walking. *Gait Posture* **59**, 134–139 (2018). <https://doi.org/10.1016/j.gaitpost.2017.10.007>

77. Wren, T.A.L., Do Patrick, K., Rethlefsen, S.A., Healy, B.: Cross-correlation as a method for comparing dynamic electromyography signals during gait. *J. Biomech.* **39**, 2714–2718 (2006). <https://doi.org/10.1016/j.jbiomech.2005.09.006>
78. Asbeck, A.T., De Rossi, S.M.M., Holt, K.G., Walsh, C.J.: A biologically inspired soft exosuit for walking assistance. *Int. J. Robot. Res.* **34**, 744–762 (2014)
79. Franz, J.R., Kram, R.: The effect of grade and speed during walking. *Gait Posture* **35**, 143–147 (2013). <https://doi.org/10.1016/j.gaitpost.2011.08.025>
80. Schmitz, A., Silder, A., Heiderscheit, B., Mahoney, J., Thelen, D.G.: Differences in lower-extremity muscular activation during walking between healthy older and young adults. *J. Electromyogr. Kinesiol.* **19**, 1085–1091 (2009). <https://doi.org/10.1016/j.jelekin.2008.10.008>

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Authors and Affiliations

Wei Jin^{1,2} · Jiaqi Liu³ · Qiwei Zhang^{1,2} · Xiaoxu Zhang^{1,2} · Qining Wang⁴ · Jian Xu³ · Hongbin Fang^{1,2}

✉ H. Fang
fanghongbin@fudan.edu.cn

- ¹ Institute of AI and Robotics, State Key Laboratory of Medical Neurobiology, MOE Engineering Research Center of AI & Robotics, Fudan University, Shanghai, 200433, China
- ² Yiwu Research Institute, Fudan University, Yiwu, Zhejiang Province, 322000, China
- ³ School of Aerospace Engineering and Applied Mechanics, Tongji University, Shanghai, 200092, China
- ⁴ College of Engineering, Peking University, Beijing, 100871, China